PANINI LINGUISTICS OLYMPIAD

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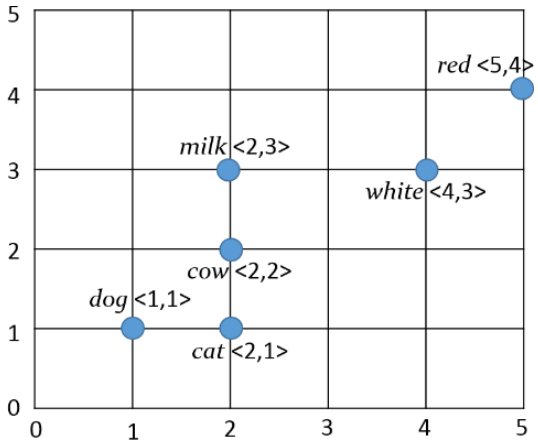
EASY PROBLEMS

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1. Word Embeddings

Problem designed by Monojit Choudhury

Computers can "compute" mathematical functions that take numbers as input and give numbers as output. Therefore, in order to make computers understand and process human language, it is useful to represent words in terms of numbers. Computer scientists have come up with sophisticated techniques that encode words as "vectors", i.e., a sequence of numbers. These are called *word embeddings.* It might be easier to think of word embedding as a mapping of words to points in an *n*-dimensional space. Let us assume that $n = 2$, i.e., words are mapped to points in a two-dimensional plane.



The diagram in the left shows hypothetical word embeddings for 6 English words. As you can see in this example, words which have similar meanings or are used in similar context have word embeddings which are close to each other, i.e., the points are closer in space. Note: The embeddings in this example are NOT related to the problem below.

Here are some English words in alphabetical order:

boy, first, girl, grammar, language, literature, man, mathematics, mathematician, number, one, position, queen, second, time, two, woman

Shown below, in no particular order, are the translations of these words in the Santali language writtenin the Ol Chiki script, and their (the English words') embeddings.

#	Santali Translation	English Embedding
A	ᱥᱟᱦᱟᱲ	<3,3>
B	ᱥᱟᱦᱟᱲᱫᱟᱨ.	<3,2>
C	ᱯᱥᱦᱟᱲ	<5,4>
D	ᱩᱞᱪ	<7,2>
E	ᱛᱟᱱ	<7,4>
F	ᱩᱞᱪᱟᱜ	<8,2>
G	ᱛᱟᱱᱫᱟᱜ	<8,4>
H	ᱢᱟᱭ᱾ᱭ	<9,3>
I	ᱢᱟᱞᱟᱜ	<10,6>

#	Santali Translation	English Embedding
J	ᱫᱟ.ᱮᱞ	<8,13>
K	ᱫᱟᱞ	<4,7>
L	ᱫᱟᱮᱟᱞ	<5,6>
M	ᱣᱟᱭᱩᱜᱩᱪ	<3,8>
N	ᱦᱟᱞᱟ	<9,10>
O	ᱦᱟᱞᱟ ᱜᱩᱪᱩ-ᱫᱟ.	<11,11>
P	ᱦᱦᱞᱞ	<8,12>
Q	ᱦᱦᱞᱞ ᱜᱩᱪᱩ-ᱫᱟ.	<10,13>

Assignment 1 [Answer Booklet]: Match the English words to their Santali translations – A to Q. No explanation needed.

Assignment 2 [Answer Booklet]: Can you guess the embedding of the words: *king*, *linguist*, *princess*? Explain your answer.

Assignment 3 [Answer Booklet]: Give the English translation of the following Santali word.



Notes: Santali is a language in the Munda subfamily of Austroasiatic languages. It is spoken by around 6.2 million people in India, Bangladesh, Bhutan and Nepal. Most of its speakers live in India, in the states of Jharkhand, Bihar, Odisha, Tripura, Mizoram, Assam and West Bengal. Since Santali did not have a script, it was written using the Roman script or the Eastern Nagari (Bangla) script. However, none of the existing scripts were able to phonetically represent the Santali language. This resulted in the invention of a new script called *Ol Chiki* by Pandit Raghunath Murmu in 1925. He is popularly known as Guru Gomke among the Santals.

Assignment 1: (6 marks)

Translate the following English sentences to Altai:

(a) The mother found this book in the school.

(b) We played on the mountain yesterday.

(c) The boy fed a horse on the lawn.

(d) You wrote a letter yesterday.

Assignment 2: (6 marks)

Translate the following Altaian sentences to English:

(a) ол ол самарады кычырды.

(b) бала менди сакды.

(c) олор кече көлде jӱсген.

(d) мен кече школыпды барганм.

Assignment 3: (8 marks)

How are words or parts of words put together to form a valid Altaian sentence? What are their meanings and functions?

– Aparna Gupte

Problem 6: Twisted Translations (20 marks)

Word embeddings are numeric representations of words as points in d-dimensional space. That is, a word can be represented using d numbers, where each number represents the coordinate of that number along a particular dimension.

For example, 2-dimensional embeddings are representations of words as points on a plane using a pair of numbers. The first number corresponds to the x-corrdinate of the point and the second number to the y-coordinate of the point.

These word embeddings are very useful representations of words for training machine learning models to perform natural langauge processing tasks. The embedding for a word is computed by looking at the word in a large corpus of text, and looking at what words occur in the context of that word.

Some machine translation models use embeddings of words in the source language to compute the embedding of the word in the target language, and then look up the word corresponding to that embedding. In this problem, we consider translating Hrangkhol to English[7].

[7]Hrangkhol or Hrangkhawl is a Kuki-Chin language spoken by the Hrangkhawl people mainly in Assam and Tripura states in India, with a minority living in Manipur and Mizoram.

Here are some English words, along with their embeddings in two dimensional space:[8]

(a) bull [2, 2]
(b) cow [3, 3]
(c) mare [4, 5]
(d) (female) duck [3, −1]
(e) son-in-law [−5, 1]
(f) daughter-in-law [−4, 2]
(g) grandson [−6, 3]

Here are some Hrangkhol words, along with their embeddings in two dimensional space:

(i) vôkchal [−4, 6]
(ii) vôkpui [−6, 8]
(iii) sakorchal [−8, 6]
(iv) arkhong [4, 2]
(v) arpui [2, 4]
(vi) vatôkhong [4, 4]
(vii) vaikikhong [8, 4]
(viii) naipa [−4, −12]
(ix) nainu [−6, −10]
(x) tunu [−8, −10]

The English words corresponding to (i-x) in **random** order are:[9]

(A) male parrot
(B) hen
(C) sow
(D) daughter
(E) boar
(F) granddaughter
(G) drake
(H) stallion
(I) son
(J) rooster

Assignment 1: (10 marks)

Match the Hrangkhol words (i-x) to their English translations (A-J).

Assignment 2: (3 marks)

Write down the word embeddings for the Hrangkhol translations of the words (a-g).

Assignment 3: (4 marks)

Write down the word embeddings for the English translations of the words (i-x).

[8] A bull is an adult male cow, and a mare is an adult female horse.
[9] A sow is an adult female pig and a boar is an adult male pig. A stallion is an adult male horse. A drake is an adult male duck. A rooster is an adult male chicken.

Assignment 4: (3 marks)

Translate the following words to Hrangkhol:

(I) female parrot

(II) he-goat (root of goat is kêl)

(III) old woman (root of old person is tar)

– Saujas Vaduguru

This is the end.
We would like to hear your feedback about this exam!
After the exam is over, please visit `http://tiny.cc/plo2020senior-feedback`.

3. The Matses Hunter Boy (20 marks)

Problem designed by Abhishek Dedhe

Deep in the Peruvian Amazon, there is a small village inhabited by members of the Matses tribe. Read the following story about a young (and slightly arrogant) Matses boy called Abhishek.

A) One fine day, Abhishek decides to go hunting in the woods. He comes across fresh footprints of wild pigs on the ground. Being an accomplished hunter, he infers that wild pigs had passed by not long ago. Shortly after that (within a few hours), he meets a group of friends who are also hunting wild pigs and says to them – "**They passed by**."

B) The next day, Abhishek decides to go hunting again. This time he comes across old footprints of wild pigs. He realizes that these are from the last hunting season. A few hours later, he meets his friends who have also decided to spend the day hunting wild pigs (again)! Abhishek tells his friends – "**They passed by**."

C) Next year, Abhishek attends a Hunting Seminar. While boasting about his skills as a hunter, he tells the Committee about the time he had discovered fresh footprints of wild pigs and how he had inferred that – "**They passed by**".

D) During the meeting, Abhishek realizes that the Seminar is being held in a hut that was very recently built by a different Native American tribe (not the Matses Tribe). Abhishek observes – "**Non-Matses Indians built a hut**."

E) After the Seminar ends, Abhishek goes to Lima, the capital of Peru. He sees a cathedral, and finds out that it was built by the Spanish during the colonial era. He exclaims – "**Foreigners built a hut.**"

F) When he returns to his village after a year, he tells his friends about the cathedral he had seen in Lima – "**Foreigners built a hut.**"

The translations of the Abhishek's sentences (A-E) are given below (in no particular order):

1. *matses-utsi bëste-wa-ak-o-ṣh*
2. *chotac bëste-wa-nëdak-onda-ṣh*
3. *chotac bëste-wa-nëdak-o-ṣh*
4. *kuen-nëdak-o-ṣh*
5. *kuen-ak-onda-ṣh*
6. *kuen-ak-o-ṣh*

Assignment 1 [Answer-Booklet]: Determine the correct correspondences.

Assignment 2 [Answer-Booklet]: Give the Matses translations of Abhishek's sentences (in bold) from the scenarios described below.

a) During the Hunting Seminar, Abhishek describes to the Committee how he was able to identify the footprints of wild pigs that were very old (from the last hunting season). He boasts that he was able to infer that – "**They passed by.**"

b) During his tour of Peru, Abhishek goes to see the Machu Picchu (an ancient architectural marvel - often counted among the Seven Wonders of the World). Having grown up in a very small village, Abhishek does not know anything about the Machu Picchu, and thus decides to read about its history from a tourist brochure. He learns that it was built by the Inca people. However, Abhishek is unimpressed, and immediately makes an arrogant comment about the Machu Picchu - "**Non-Matses Indians built a hut**."

c) After touring Peru for more than a year, Abhishek returns to his village. He is dismayed to see that his lavish mansion had fallen into ruins. In its place is a small, shabby hut. Abhishek is told that this hut was built by the kind-hearted locals just a few days ago. Humbled and ashamed, Abhishek falls to his knees and cries out – "**Matses Indians built a hut.**"

Assignment 3: Translate this sentence to English: "Abhishek bëste-wa-nëdak-onda-şh". Let your imagination run wild and come up with an appropriate context in which you might utter this sentence! (Try to avoid ambiguous scenarios).

Assignment 4: Explain your solution.

Problem no 4. Tangkhul Adjectives

Study the following sentences or phrases from Tangkhul-Naga language, spoken in Manipur, India.

1.	Kachuia shinaao	Tall woman
2.	Katyeoa shim	Small house
3.	Kaphaa lairik	Good book
4.	Kathi thingrong	Dead tree
5.	Kathum	Three
6.	Phangaa	Five
7.	Shinaao-hi chuia	This woman is tall.
8.	Hi shim-phangaa na	There are five houses.
9.	Hi kachuia shinaao na	This is a tall woman.
10.	Kachui	Tall
11.	Kaphaa	Good
12.	Katyeo	Small
13.	Kathi	To die/ Dead
14.	Thingrongping thi dalai	Trees are dying.
15.	Thingrong-chi thira	That tree will die.
16.	Thingrong-chi chuira	The tree will be tall.
17.	Kashung khamong	The closed door
18.	Aa-na khamong shunga.	He closed the door.
19.	I-na khamong-chi shungra.	I will close the door.
20.	Hi kaphangaa lairik na.	This is the fifth book.
21.	Chi kakathuma shim na.	That is the third house.
22.	I-na lairik pama	I wrote a book.
23.	Aa-na lairik pam delai	I am writing a book.

A] Translate the following in Tangkhul:

1. The good house.
2. He wrote five books.
4. He is writing the fifth book.
5. The small woman
6. The small woman closed the second door.
7. He will be small.
8. That is a dead tree.
9. This tree is dead.
10. This door is closed.

B] Can you find any pattern in the forms that the adjectives and verbs of Tangkhul-Naga language take according to where they occur in a sentence? Explain the pattern, if there is any. What can you say about the categories of verbs and adjectives in this language?

Problem 4: No Zeroes in Xerénte (14 marks)

Given below are some number phrases in Xerénte[1] and their equivalents in Hindu-Arabic numerals, in random order. All the numbers in this problem are greater than zero and less than a hundred.

sahure-pra'nẽ, smĩ'si, mrẽpränẽ, sikwa're, srure-'sru, sahure-si'ptõ,
siptõ're, pranẽ're, sikwaĩ'pse, sipta'tõ, sahure-si'kwa, ponkwa'nẽ

1, 2, 3, 4, 5, 13, 14, 15, 30, 40, 50, 77

Here are some hints:

- The difference between sru'ru and ponkwa'nẽ lies between sikwaĩ'pse and sipra'nẽ.
- sikuiwa'mẽ + sikwaĩ'pse = ĩssui'bka + mrẽpränẽ

Assignment 1: (4 marks)

Match the Xerénte number phrases to their equivalent numbers.

Assignment 2: (2 marks)

Write these numbers in Xerénte:

(a) 11
(b) 32
(c) 66
(d) 89
(e) 98

Assignment 3: (2 marks)

Write these numbers in Hindu-Arabic numerals:

(a) kuburere-'bka
(b) sikware-kui'wa
(c) pranẽre-si'nẽ
(d) kuburere-kwa'nẽ
(e) pranẽre-mĩ'si

Assignment 4: (6 marks)

How are words or parts of words put together to form numbers in Xerénte? What are the functions of these words or parts?

– Mukund Choudhary

[1]The Xerénte language (also known as Sherenté) is spoken by the Xerénte people of Brazil.

Problem #3: Mnemonic Major System (15 marks)

The Mnemonic Major System is a technique used by mental athletes who take part in memory championships. This technique is used to convert numbers to meaningful words that are easy to visualize, and hence easy to memorize. Listed below are a few examples.

314,159,265,358: **meteor tail banshee lime loaf**

75,141,195: **cluttered table**

701,894: **ghost vapour**

99,501,247: **pebble stone rock**

7,512,026: **golden snitch**

8: **hive**

854: **flower**

2,394,429**: number rainbow**

6,096**: cheese bush**

Note: The commas are not related to the encoding. They just make the numbers easy to read.

Convert these phrases into numbers:

a) **short striped zebra**
b) **moon cash**
c) **cat dog**
d) **grey elephant**

The earliest version of the Mnemonic Major System was designed by Pierre Hérigone (1580–1643), a French mathematician and astronomer. The name "major system" refers to Major Beniowski, who published a version of the system in his book, "*The Anti-Absurd or Phrenotypic English Pronouncing and Orthographical Dictionary*".
Similar number memorization systems also occur in other parts of the world, such as the *Katapayadi* system going back to at least the 7th Century AD in India.

- Anindya Sharma

Problem #5: Japanese Compound Words (20 marks)

In Japanese, certain root words undergo a change at their left edge when they form compound words under specific circumstances. This phenomenon is traditionally called *Rendaku*. Examine the following examples of Japanese compound words:

Japanese words	Compound word	Gloss	Meaning
hito* + *hito	***hitobito***	person + person	people
ike* + *hana	***ikebana***	keep alive + flower	flower arrangement
toki* + *toki	***tokidoki***	time + time	sometimes
te* + *kami	***tegami***	hand + paper	letter
hitori* + *tabi	***hitoritabi***	one person + travel	travelling alone
tsuno* + *tokage	***tsunotokage***	horn + lizard	horned lizard
yama* + *kawa	***yamakawa***	mountains + rivers	mountains and rivers
aka* + *huda	***akahuda***	red + tag	red tag
maki* + *susi	***makizusi***	rolling + sushi#	rolled sushi
yama* + *kawa	***yamagawa***	mountain + river	mountain's river
hebi* + *tosi	***hebidosi***	snake + year	snake year
eda* + *ha	***edaha***	branches + leaves	branches and leaves
oo* + *kaze	***ookaze***	big + wind	big wind
kokoro* + *tsukai	***kokorodzukai***	heart + using	anxiety
denki* + *kurage	***denkikurage***	electric + jellyfish	electric jellyfish
yuki* + *kaeri	***yukikaeri***	coming + going	coming and going
hana* + *sono	***hanazono***	flower + garden	flower garden
nise* + *karasu	***nisegarasu***	fake + crow	fake crow

Assignment 1:
Fill in the following compound words.

Japanese words	Compound word	Gloss	Meaning
wa* + *kasi	_____	Japanese + cake	Japanese cake
yama* + *kazi	_____	mountain + fire	forest fire
juzu* + *tsunagi	_____	rosary + sequence	roping together
yama* + *tera	_____	mountain + temple	mountain temple
ao* + *kawazu	_____	green + frog	green frog
uwa* + *huta	_____	top + lid	top lid
e* + *tako	_____	picture + kite	picture kite

Assignment 2:
When compounds are made up of more than two words, things become truly interesting. Fill in the blank entries (A to E) in the following Table:

Japanese words	Compound Word	Gloss	Meaning
hosi-kaki-tsukuri	***hosi-gaki-dzukuri***	dried persimmon* making	making dried persimmon
nise-kami-tana	***nise-kami-dana***	fake divine altar	a divine altar that is fake
nise-kami-tana	***nise-gami-dana***	fake divine altar	an altar for a fake God
ori-kami-tana	***ori-gami-tana***	fold paper shelf	origami shelf
nise-sakura-dayori	***A. ?***	fake cherry-blossom tidings	tidings of fake cherry-blossoms
nise-sakura-dayori	***nise-sakura-dayori***	fake cherry-blossom tidings	fake tidings of

			cherry-blossoms
nuri-hasi-hako	***B. ?***	lacquered chopstick box	a box for lacquered chopsticks
nuri-hasi-hako	***nuri-hasi-bako***	lacquered chopstick box	a lacquered box for chopsticks
nise-tanuki-siru	***C. ?***	pseudo tanuki** soup	soup made from pseudo tanuki
nise-tanuki-siru	***D. ?***	pseudo tanuki soup	tanuki soup that is not authentic
oya-ko-kenka	***E. ?***	parent child quarrel	parent and child quarrel

Japanese is spoken by about 125 million speakers primarily in Japan.
#**sushi** is cooked vinegared rice combined usually with raw fish.
***persimmon** is a fruit.
****tanuki** is Japanese raccoon dog.

- *Zarana Sarda*

Problem no 5. [20 marks]

On the basis of the data given below, explain the differences in question formation process in the languages mentioned.

[Abbreviations: CM = Sentence completion marker; QM: Question marker; TOP: Topic particle]

1) Suma reads 'Crime and Punishment'. (English)
   What does Suma read?

2) Suma 'Crime and Punishment' sim e. (Simte)
   Suma 'Crime and Punishment' read CM

   Bang tangthubu Suma sim ei?
   Which novel Suma read QM

3) Suma "Crime and Punishment" waacate (Marathi)
   Suma Crime and Punishment reads

   Suma kaay waacate?
   Suma what reads

4) Sumawa "Crime and Punishment" to iu hono yomi masu. (Japanese)
   Suma-TOP "Crime and Punishment" thus called book reads-CM.

   Sumawa nan to iu hono yomi masu ka.
   Suma what thus called book reads QM

Problem #2: Valman (15 marks)

The English conjunction *and* is translated into the *Valman* language differently depending on the words it connects. Consider some English phrases, and the corresponding forms of the conjunction *and* which will appear in the *Valman* translations of these phrases:

me and you	**mcha**
me and my brother	**man**
you and the chieftain	**nan**
you and your neighbours	**nay**
the neighbour and I	**npa**
the chieftain and his wife	**na**
the neighbour's mother and I	**wpa**
my wife and the guests	**way**
your sisters and you	**ycha**
my brothers and the chieftain's mother	**ya**

Valman or *Walman* is a language of the *Torricelli* language family spoken by less than 2000 people in Papua New Guinea.

y and **ch** are vowels.

Assignment:

Predict the form of the conjunction *and* in Valman translations of the following English phrases? Explain your solution.

the chieftain and you, the guest and the hosts, your neighbours and I, you and the guests' wives, my brother and the chieftain, me and the chieftains, grandmother and my wife

- P. Arkadiev

Problem #3: Translating Welsh (15 marks)

Machine Translation is a very important technology on which computer scientists and linguists have been working together for many years now. The aim of this technology is to develop computer systems that can automatically translate sentences from one language to another. Though machine translation is a very difficult problem, there are systems that can solve this problem reasonably well. Given below are 8 input English sentences and their corresponding output Welsh translations generated by one such well-known machine translation system developed by *Google*.

1. The man kills the ants.
   Mae'r dyn yn lladd y morgrug.
2. The ant kills.
   Mae'r morgrugyn yn lladd.
3. The men have killed.
   Mae'r dynion wedi lladd.
4. The dog jumped on the man.
   Mae'r ci yn neidio ar y dyn.
5. The ants will be killed.
   Bydd y morgrug yn cael eu lladd.
6. The dog was going to kill.
   Roedd y ci yn mynd i ladd.
7. The man is going to jump on the ant.
   Mae'r dyn yn mynd i neidio ar y morgrugyn.
8. The dog will be sent.
   Bydd y ci yn cael ei anfon.

Welsh is the native and official language of Wales. It has almost 700,000 speakers, mostly from England and Wales.

a, e, i, o, u, w and ***y*** are vowels in Welsh. **y** is pronounced as "i" in "bit"; **w** is pronounced as "oo" in "book"; ***c*** is pronounced as in "case", ***dd*** is pronounced as "th" in "this", whereas ***d*** is pronounced as in "day"; similarly, ***l*** and ***ll*** are pronounced differently. An apostrophe (as in ***Mae'r***) indicates a stress on a syllable, and a hat (e.g., ***ŵ***) indicates a long vowel.

Assignment 1:

What do you think would be the output of this translation system for the following English sentences?

a) The ants have killed the man.
b) The dog jumps on the men.
c) The men were going to jump on the ants.

Assignment 2:

Can you guess the English input sentences to the machine translation system that led to following Welsh translations? If you think that it is not possible to guess the exact English input for some of the words or sentences, then explain why you think so, and what the possible inputs could be.

d) ***Mae'r ci yn lladd.***
e) ***Mae'r dyn yn anfon SMS.***
f) ***Mae'r cŵn yn mynd i gael ei anfon.***

– Prachatos Mitra

Problem #1: Solving Lakota Equations (15 marks)

Here are some equations written in Lakota involving only positive natural numbers:

wanji* × *nunpa* = *nunpa
yamni* + *nunpa* = *zaptan
yamni* + *topa* = *nunpa* + *zaptan
šakpe* + *topa* = *nunpa* × *zaptan
nepcunka* – *šakpe* = *yamni
wikcemna* × *ake wanji* = *opawing'e sam wikcemna
wikcemna yamni* + *wanji* = *ake wanji* + *wikcemna nunpa
šaglohan* – *topa* = *topa

Lakota is a Native American language spoken in the United States by approximately 6,000 speakers living mostly in the northern plains states of North Dakota and South Dakota.

One of the interesting features of the Lakota language is that men exclusively speak certain words, and similarly there are words that are only used by women.

Assignment 1:

Complete the following equations in Lakota.

a) ***wikcemna* + *nunpa* = _____**

b) ***ake wanji* + *wanji* = *nunpa* + _____**

c) ***wikcemna yamni* = *zaptan* × _____**

Assignment 2:

Which numbers do the following Lakota expressions represent?

d) ***ake topa***

e) ***opawing'e sam yamni***

Assignment 3:

Express the following numbers in Lakota?

f) 13

g) 20

– Nilai Sarda

Problem No. 5 (15 Points):

Amharic

Consider the following English sentences and their translations into the Amharic language written out in Latin transcription:

1. She tastes coffee, not juice.	bunna ənǧi čəmmaḳi atḳämsəm
2. You sell a horse, not a mule.	färäs ənǧi bäḳlo atšäṭum
3. She rides a mule, not a camel.	bäbäḳlo ənǧi bägəmäl athedəm
4. You like fish.	asa təwäddalläh
5. She fries chicken.	doro təṭäbsalläč
6. You do not taste meat.	səga atḳämsum
7. He goes by train.	bäbabur yəhedal
8. They sell a chicken.	doro yəšäṭallu
9. You choose a car.	mäkina təmärṭallačhu
10. They taste meat, not fish.	səga ənǧi asa ayḳämsum

Assignment 1. Translate into English:

1. asa təḳämsalläh
2. čəmmaḳi təšäṭallačhu
3. färäs aymärṭum

Assignment 2. Translate into English in all possible ways:

bäbabur ənǧi bämäkina athedəm

Assignment 3. Translate into Amharic:

1. They like fish, not meat.
2. You ride a horse.
3. He does not fry a chicken.
4. She chooses a camel.

Note. *š* is pronounced similarly to English *sh* as in *show*, *č* is pronounced similarly to English *ch* as in *cheap*, *ə* and *ä* are specific vowels of the Amharic language, *ǧ*, *č*, *ṭ* and *ḳ* are specific consonants.

[1] Amharic is a Semitic language spoken by more than 17 million people in Ethiopia.

-M. Bulakh

3. Yash in Rwanda (20 marks)

Problem designed by Yash Sinha

Yash, who is studying linguistics in the University of Chicago, went to Rwanda on a field trip. His task was to analyze the Kinyarwanda language, which is one of the three official languages of Rwanda along with French and English. What flummoxed him though is while some English sentences had only one translation in Kinyarwanda, some had two and some even more. To crack the mystery, Yash collected the Kinyarwanda translations of a few English sentences from the locals, though he could not make his subjects speak out all the possible translations (after all they were not linguists and said whatever came to their mind). Yash prepared the following table of Kinyarwanda sentences and their English sentences and also made some notes of how many translations he thought the English sentence actually had for cases where he did not get all possible translations.

1.	*Umugabo arakora*	The man is working.
2.	*Abogore baragenda*	The women are going.
3.	*Umugabo arabona umugore*	The man is seeing the woman.
4.	*Umugore arabonwa numugabo*	
5.	*Umuhungu arabonwa nabogore*	The women are seeing the boy.
6.	*Abogore barabona umuhungu*	
7.	*Abogabo baraandika ibaruwa nikarumu*	The men are writing the letter with the pen. [There are 2 more possible translations in Kinyarwanda]
8.	*Abogabo baraandikiisha ikarumu ibaruwa*	
9.	*Umuhungu arabona abogore nijisho*	The boy is seeing the women with the eye. [there are 2 more possible translations in Kinyarwanda]
10.	*Ijisho araboniishwa abogore numuhungu*	
11.	*Umualimu araandikera umuhungu igitabo*	The teacher is writing the book for the boy. [there are 2 more possible translations in Kinyarwanda]
12.	*Umualimu araandika igitabo kuumuhungu.*	

Yash realized that for every pair of Kinyarwanda sentences that mean the same thing, the two forms mean the same thing. *However, they do not necessarily have the same structure.*

Assignment 1: Explain how the same meaning can be expressed in multiple ways.

Assignment 2 [Answer-booklet]: Translate the following sentences into English

a) *Ibaruwa araandikwa nabogabo*
b) *Ibaruwa araandikwa numugore kuabogabo*
c) *Abogabo barabonera umugore abohungu*

Assignment 3 [Answer-booklet]: Now translate the two sentences below. Consider the verb form carefully.

d) *Ikarumu araandiishwa ibaruwa numagabo*
e) *Abogore barabonerwa abohungu numugabo*

Assignment 4 [Answer-booklet]: Provide Kinyarwanda translations for these two English sentences. See instructions besides each sentence for the number of translations you are expected to provide.

f) The boys are writing the book. (Give ***two*** Kinyarwanda equivalents)
g) The boy is writing the book for the women. (Give ***four*** Kinyarwanda equivalents)

Problem no 3. Simte Verbs

Following is the data given from Simte language. Simte is a language of the Kuki-Chin group spoken in southern parts of Manipur.

Ka puanaahtung lian ahi = my coat is big.
Guul lian = big snake
Kei puanaahtung silh = I am wearing a coat
Amah lalluh khuh paat= He is wearing a crown
Ka puan asan rong ahi = my dress is red in colour
Kei puan teng = I am wearing a dress
Chi puanaahtung asang rong ahi = this coat is red in colour
Na khi bangtan akhat ahi = your necklace is pretty
Ka puan bangtan akhat ahi = my dress is pretty
Nang puannah silh = you are wearing a shirt
Amah puanten teng = she is wearing a puanten (a traditional wear worn like a wrap-around skirt by the women)
Chi lukhuh maidawh aeng rong ahi = this cap is yellow in colour
Kei khi awh = I am wearing a necklace
Chi kengtophah lian ahi = these shoes are big
Kei kengtophah deih = I like shoes
Kei kengtophah but = I am wearing shoes
Nang tau but = you are wearing bangles
Amah tokatual teng = he is wearing pants
Samuldial akang ahi = the muffler is white in colour
Nang samuldial awh = you are wearing a muffler
Lalluh khuh neo ahi = the crowns are small
Amah lukhuh maidawh paat = she is wearing a cap

Q1. Translate the following:

1) He is wearing a big white muffler
2) I like necklaces
3) Your dress is yellow in colour
4) She is wearing a red bangle
5) He is wearing this small cap.

Q2. Match the following:

1) Samul puanaah (sweater)	a) teng
2) Bil bah (earring)	b) silh
3) Ngwang tuamna dial (scarf)	c) but
4) Lukhuh (hat)	d) paat
5) Tokatual (trouser)	e) awh

5. Washo (20 marks)

Problem designed by Yash Sinha

Consider the following sentences from Washo, and their English translations.

(1)	*teliwhu basa' basa'ige da'mo'mo' sili*	The man gave to the woman the book that he wrote.
(2)	*basa' digyisge teliwhu disili*	The man gave to me the book that I saw.
(3)	*da'mo'mo' basa' basa'isge dipigelyayi*	I returned to the woman the book that she wrote.
(4)	*basa' umbasa'ige teliwhu umsili*	You gave to the man the book that you wrote.
(5)	*teliwhu basa' umsilisge dipigelyayi*	I returned to the man the book that he gave to you.
(6)	*basa' umgyige da'mo'mo' umsili*	You gave to the woman the book that you saw.
(7)	*teliwhu da'mo'mo' basa' silige gyi*	The man saw the book that he gave to the woman.

Assignment 1 [Answer Booklet]**:** Translate the following sentences into English. If there are multiple translations possible, give all of them.

a. *da'mo'mo' basa' umsilige basa'i*

b. *teliwhu basa' disilisge digyi*

c. *da'mo'mo' basa' disilisge dipigelyayi*

Assignment 2 [Answer Booklet]**:** Translate the following sentences into Washo.

a. The man gave to me the book that I returned to him.

b. You saw the book that you gave the woman.

Assignment 3: Explain how the language works by describing the order in which words are used in a sentence, the translations for different verbs and nouns, the way in which verbs and nouns get modified and any other things that you find interesting about the language.

Notes: (1) Washo is an endangered Native American language isolate spoken by the Washo on the California–Nevada border in the drainages of the Truckee and Carson Rivers, especially around Lake Tahoe. While there are only 20 elderly native speakers of Washo, since 1994 there has been a small immersion school that has produced many moderately fluent younger speakers. This problem is based on data from Emily Hanink's field work. (2) Some of the words in this problem have been altered slightly for ease of solvability; for the same reason, diacritics have been omitted.

Problem No. 2 (10 Points):
Vietnamese

Consider the names of some European states written in Vietnamese and their **unordered** English translations:

Áo, Ba Lan, Bỉ, Đan Mạch, Hà Lan, Na Uy, Phần Lan, Thụy Điển, Thụy Sỹ, Ý.

Italy, Austria, Denmark, Norway, Switzerland, Netherlands, Belgium, Sweden, Finland, Poland.

Assignment 1. Match the Vietnamese country names with the corresponding English translations.

Assignment 2. What can you say about the origin of the name *Hà Lan*?

Note. *â* and *ê* are pronounced as *e*, *y* is pronounced as *i*, *đ* is pronounced as *d*, *ph* is pronounced as *f*, *ch* is pronounced similarly to English *k* as in *keep*, *uy* is pronounced similarly to English *we* or French *oui*. The diacritic symbols ´,`,̉,̃ mark specific ways of pronouncing the vowels (so called tones).

- A. Somin

Problem #4: Turkish News and Conversations (15 marks)

Read this account of some interactions between three Turkish speakers: A, B and C. The account is mostly in English, with Turkish sentences quoted verbatim in some places.

> A had told B that the king was sick, so B said to C, "**Kral hastaymış.**" To this C objected and said it was not the king who was sick, but the minister. He had seen that the minister was sick. So C said, "**Bakan hastaydı.**" After this, C goes home, and A joins B.
>
> A and B talk about the town center which was demolished. A is too young to have seen the town center so B tells him that it was old: "**Şehir merkezi eskiydi,**" he said. After that, B invited A to his house on the 18th of May, but he said he wasn't available that day, but he was available a day before that, "**17 Mayıs'ta müsait değilim**". B also went and invited C.
>
> When A, B and C met on 17th May, A told C, "**Şehir merkezi eskiymiş**" and also that the minister is recovering now: "**Bakan şu an iyileşiyor**". He further said that the minister was seen earlier, "**Daha önce görüldü**"

Now, consider small excerpts from two news reports in Turkish. They are reporting the same event(s). One of these reports is based on the other one, i.e., one of them was written earlier and the other report used the original one as a basis. Your task is to determine which is the original report. As always, justify your answer. Also, translate as much of the reports as you can. There will be portions which you cannot translate; you can put ellipses (...) for them and translate whatever else you can.

Report 1:

> **Bakan yorum yapmaya müsait değilmiş. Daha önce hastaymış, şu an iyileşiyor. 12 Mart'ta şehir merkezinde görülmüş.**

Report 2:

> **Bakan en son 12 Mart'ta şehir merkezinde görüldü. Gazetemiz ona ulaşmaya çalıştığında yorum yapmaya müsait değildi. Daha önce hastaydı, şu an iyileşiyor.**

Turkish belongs to the Turkic language family. It is spoken in Turkey and within smaller communities in many other countries. It is the mother tongue of more than 63 million people.
Note that i and ı are distinct letters.

– Yash Sinha & Atınç çağan şengül

Problem No. 4 (25 Points):
Tiwa

Consider the following sentences in the Southern Tiwa language[1] and their English translations:

kakhwianmuban	I saw your dog.
kamkeuapthãban	I found your boot.
kampimikiban	I fed your deers.
kanide amkahunmuban	You saw the horse's box.
piide owkeuaptuwiban	You bought the deer's boots.
seuanide amkantuwiban	You bought the man's horses.
seuanide towmukhithãban	I found the man's hats.
'uide takhwianmikiban	I fed the child's dog.

Assignment 1. Translate into English:

kowkahuntuwiban
'uide tamkhwianthãban

Assignment 2. Translate into Southern Tiwa:

I bought the dog's hat.
I saw your boots.
You fed the man's child.

Note. *ã* is a specific vowel and *'*, *kh* and *th* are specific consonants of the Southern Tiwa language.

[1] The Southern Tiwa language belongs to the Kiowa-Tanoan language family. It is spoken by 1500 people in the U.S. state of New Mexico.

- P. Arkadiev

Problem No. 1 (5 Points):
Norwegian

Consider four words in the Norwegian language[1] :

farmor, farfar, mormor, morfar

The first one can be translated into English as ‘grandmother’, but good Norwegian-English dictionaries usually give the more precise translation.

Assignment. Find out the precise meaning of the Norwegian word farmor. Translate the remaining three words into English. Explain your solution.

[1] The Norwegian language is a North Germanic language spoken primarily in Norway, where it is the official language. It is spoken by more than 5 mln people.

- Yu. Kuznetsova

Problem 2: Stressed Manobo (15 marks)

Given below are some words in Sarangani Manobo[2]. Each word has the location of stress (i.e. emphasis) marked with ˈ at the beginning of the stressed syllable.

ˈbaso, deˈitek, meneˈnoo, meˈama, beˈgas, leˈkat, ˈotaw,
eteˈbay, bineleˈsan, ˈdeget, ˈbenget, miˈneles, ˈikan, ˈdoen

Assignment 1: (10 marks)

Mark the stress in the following Sarangani Manobo words with ˈ at the beginning of the stressed syllable:

(a) mengabat
(b) migbasa
(c) iselem
(d) benal
(e) tadon
(f) belegkong
(g) mola
(h) medaet

Assignment 2: (5 marks)

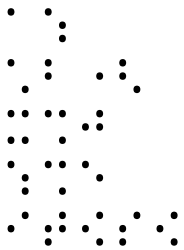
Given a new word in Sarangani Manobo, how would you identify the syllables in the word? Once you have identified the syllables, how would you determine which syllable is stressed?

– Saujas Vaduguru

Problem 3: Tenji Braille (15 marks)

Here are five Japanese words and their corresponding Tenji Braille[3] transliterations:

asa	morning
kaigi	meeting
fukuro	pocket
sakura	cherry blossom
otokonoko	boy



[2]Sarangani Manobo is one of the languages comprising the Manobo subgroup within the Philippine branch of theAustronesian language family. It is spoken by an estimated twenty thousand people living on the east coast of the Sarangani Peninsula and another ten thousand people living on the San Agustin Peninsula of southern Mindanao.

[3]Japanese Braille is the braille script of the Japanese language. It is based on the original braille script, though the connection is tenuous. In Japanese it is known as *tenji*, literally “dot characters”.

Here are some more Japanese words and their Tenji Braille transliterations, in no particular order:

1.	ikimono	living creatures	A.
2.	nezumi	mouse	B.
3.	sora	sky	C.
4.	shinzo	heart	D.
5.	ushiro	behind	E.
6.	hi	fire	F.
7.	kinoko	mushroom	G.
8.	mugi	wheat	H.
9.	bangohan	supper	I.
10.	mame	beans	J.

Assignment 1: (10 marks)

Match the Japanese words (1-10) to their Tenji Braille transliterations (A-J).

Assignment 2: (5 marks)

Transliterate the following words to Tenji Braille:

i.	geri	diarrhea
ii.	kirin	giraffe
iii.	saba	mackerel
iv.	hotaru	firefly
v.	atsumaru	to gather

– Apoorva Bhagwat and Sandipan Dandapat

Problem #4: Telling Time in Swahili (15 marks)

Here are eight time expressions in English and their **unordered** Swahili translations.

Sunday, 1:00 am	***jumamosi, saa moja usiku***
Sunday, 7:30 am	***jumapili, saa tatu na robu asubuhi***
Sunday, 9:15 am	***jumamosi, saa saba usiku***
Tuesday, 12:15 pm	***jumamosi, saa mbili na robu usiku***
Tuesday, 11:30 pm	***jumanne, saa tano na nusu usiku***
Saturday, 10:30 am	***jumanne, saa sita na robu asubuhi***
Saturday, 7:00 pm	***jumamosi, saa nne na nusu asubuhi***
Saturday, 8:15 pm	***jumapili, saa moja na nusu asubuhi***

Assignment 1:
Match the translations. Justify your solution.

Assignment 2:
Translate the following Swahili times into English.
jumatano, saa moja na robu asubuhi; ***jumapili, saa nne na nusu asubuhi***

Assignment 3:
Translate the following English times into Swahili
Monday, 12:15 am; Monday, 10:00 pm

The Swahili language, or Kiswahili, is a Bantu language spoken by various communities inhabiting Tanzania, Kenya, Uganda, Rwanda, Burundi, Mozambique and the Democratic Republic of Congo. Around five million people speak Swahili as their mother tongue.

– Nilai Sarda

Problem 2: Stressed Manobo (15 marks)

Given below are some words in Sarangani Manobo[2]. Each word has the location of stress (i.e. emphasis) marked with ˈ at the beginning of the stressed syllable.

ˈbaso, deˈitek, meneˈnoo, meˈama, beˈgas, leˈkat, ˈotaw,
eteˈbay, bineleˈsan, ˈdeget, ˈbenget, miˈneles, ˈikan, ˈdoen

Assignment 1: (10 marks)

Mark the stress in the following Sarangani Manobo words with ˈ at the beginning of the stressed syllable:

(a) mengabat
(b) migbasa
(c) iselem
(d) benal
(e) tadon
(f) belegkong
(g) mola
(h) medaet

Assignment 2: (5 marks)

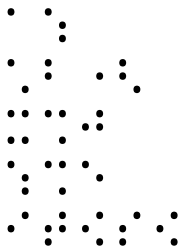
Given a new word in Sarangani Manobo, how would you identify the syllables in the word? Once you have identified the syllables, how would you determine which syllable is stressed?

– Saujas Vaduguru

Problem 3: Tenji Braille (15 marks)

Here are five Japanese words and their corresponding Tenji Braille[3] transliterations:

asa	morning
kaigi	meeting
fukuro	pocket
sakura	cherry blossom
otokonoko	boy



[2]Sarangani Manobo is one of the languages comprising the Manobo subgroup within the Philippine branch of theAustronesian language family. It is spoken by an estimated twenty thousand people living on the east coast of the Sarangani Peninsula and another ten thousand people living on the San Agustin Peninsula of southern Mindanao.

[3]Japanese Braille is the braille script of the Japanese language. It is based on the original braille script, though the connection is tenuous. In Japanese it is known as *tenji*, literally “dot characters”.

1. Sixteen Japanese Phrases (20 marks)

Problem designed by Abhishek Dedhe

Given below is a list of 16 Japanese phrases along with their English translations, in random order.

niko tamago, gohon kawa, hachihon gen, goko ringo, kyuhiki inu, gomai pureto, nihiki neko, rokumai enberopu, gohiki neko, rokumai pureto, nihon hashi, kyuhon kawa, kyumai kitte, hachihiki ari, kyuko tama, rokuko tama

two eggs, five apples, five plates, two wildcats, nine balls, five rivers, nine rivers, nine dogs, six envelopes, eight ants, nine postage stamps, two chopsticks, eight strings, five wildcats, six plates, six balls

Assignment 1 [Answer Booklet]: Match the translations.

Assignment 2 [Answer Booklet]: Given your current knowledge and that *tiger* translates to *tora* in Japanese, which of the following would be valid and why?

nihiki tora, niko tora, nihon tora, nimai tora

Assignment 3 [Answer Booklet]: Actually, none of the above four phrases are valid. Instead, you might expect to see *nito tora*. You might also see *nito zo* (*zo* = *elephant*), and *nito raion* (*raion* = *lion*).

Which of the following are correct/valid Japanese phrases? Provide the English translations for the correct ones. Note: *yajirushi* = *arrow*, *hachi* = *bee*.

hachimai hashi, rokuhon yajirushi, gohiki zo, niko ringo, hachiko hashi, goto neko

Notes: Japanese, also known as *Nihongo*, is an East Asian language spoken by about 126 million people, primarily in Japan, where it is the official language and national language. It is a member of the Japonic (or Japanese-Ryukyuan) language family, and its relation to other languages, such as Korean, is debated. Japanese has been grouped with language families such as Ainu and Austroasiatic, but none of these proposals has gained widespread acceptance. Japanese has no genetic relationship with Chinese, but it makes extensive use of Chinese characters, or *kanji*, in its writing system, and a large portion of its vocabulary is borrowed from Chinese.

Problem 6: Semantic Atlas (13 marks)

Have you ever wondered about how language might be represented in our brain? Recent work done at the University of California, Berkeley found that words with similar meanings or connotations seem to be represented in similar regions of the brain (called "voxels" which are like two-dimensional "pixels" but in three dimensions), in the form of high activity in those regions when those words are processsed. Given below are some words some brain activity patterns in five distinct voxels.

Word	**Voxel 1**	**Voxel 2**	**Voxel 3**	**Voxel 4**	**Voxel 5**
(A)	high	high	high	low	low
(B)	low	high	low	high	low
(C)	high	high	low	high	low
(D)	low	high	high	high	high
(E)	high	low	low	low	low
(F)	high	low	low	high	low
(G)	low	high	low	low	high
(H)	low	low	low	low	high
(I)	low	low	high	low	high
(J)	high	low	high	low	low
insult	low	high	high	low	high
illusion	(1)	(2)	low	(3)	(4)
murder	low	(5)	(6)	low	(7)
blood	high	high	(8)	(9)	(10)
helpful	(11)	(12)	high	(13)	(14)
belief	(15)	(16)	low	(17)	(18)

Assignment 1: (8 marks)

Match words from the following list to the appropriate spaces labelled (A)–(J) above:

fracture, stripes, depression, philosophy, torture, cut (wound/injury),
eyes, love, war film, telephone

Assignment 2: (5 marks)

Fill in the spaces numbered (1)–(18) above.

– Abhishek Dedhe

Problem 1: Sanskrit Months (15 marks)

The traditional Indian calendars, collectively known as the Hindu calendars, follow a luni-sidereal system of time keeping, where the movement of moon against the fixed background of stars is used as the basis for defining years, months and dates. While the Hindu calendars followed in various regions of India today vary greatly from each other, one thing that is common to all of them is the names of the 12 months. The Sanskrit names for these months, in the temporal order, are:

Caitra, Vaiśākha, Jyeṣṭha, Āṣāḍha, Śrāvaṇa, Bhādrapada, Āśvina, Kārtika, Mārgaśirṣa, Pauṣa, Māgha, Phālguna

The month of *Caitra* coincides with mid-March or mid-April in most of the modern Hindu calendars. Interestingly, these names have been systematically derived from the names of the *nakṣatra's* (stars or constellations) that coincide with the moon in that month on the full-moon night (*Pūrṇimā*). There are 27 *nakṣatra's* corresponding to the 27 days of the lunar cycle.

Given below are the names of 23 of the 27 *nakṣatra's*, in the right order.

Aśvinī, Bharaṇī, Kṛttikā, Rohiṇī, Mṛgaśirṣa, Ārdrā, Punarvasu, Puṣya, Aśleṣā, Pūrva Phalguṇī, Uttara Phalguṇī, Hastā, Svāti, Viśākhā, Anurādhā, Jyeṣṭha, Mūla, Śravaṇa, Śraviṣṭhā, Śatabhiṣak, Pūrva Bhādrapadā, Uttara Bhādrapadā, Revatī

It is possible to guess the names and positions of the four missing *nakṣatra's* if you use the following information:

1. All of them end with *ā*.
2. All of them begin with a short vowel. (Not really, but you can assume so.)
3. There is no missing *nakṣatra* between *Jyeṣṭha* and *Mūla*

Assignment 1:
Assuming *Aśvinī* to be the first *nakṣatra* (and *Revatī* the 27th), write down the positional numbers and the names of the 4 missing *nakṣatra's* in the list.

Assignment 2:
Write down the names of the 12 *nakṣatra's* with whom the full moon coincides for each of the 12 months.

Assignment 3:
The following 4 *nakṣatra's* do not lend their names to any month:

Hastā, Punarvasu, Śatabhiṣak, Revatī

However, if one had to derive names of months from them, what would they be? In case it is not possible to predict the month name for some of these *nakṣatra's*, write "*NP*" in the Answer Booklet.

Explain your reasoning, and then make your best guess for what the month name might be.

1. *h* indicates that the preceding consonant is aspirated (pronounced with a puff of air); *ṇ, ṣ, ṭ* denotes *n*, *sh* and *t* in *barn, marsh* and *art*, uttered with the tip of the tongue turned back (also colloquially referred to as the hard consonants); *ś* is pronounced as *sh* as in *shoe* and *s* is pronounced as *s* in *sea*; *ṛ* is a vowel; a bar over a letter (e.g., *ā, ī*) denotes a long vowel. **2.** *Sanskrit* is the primary sacred language of Hinduism, a philosophical language in Buddhism, Hinduism and Jainism, and a literary language that was in use as a lingua franca in Greater India. It is a standardized dialect of *Old Indo-Aryan*, originating as *Vedic Sanskrit* and tracing its linguistic ancestry back to *Proto-Indo-Iranian* and *Proto-Indo-European*. Today, it is listed as one of the 22 scheduled languages of India. **3.** The lunar cycle is of 27 days and 7¾ hours; the 27 days are covered by the 27 *nakṣatra's* and the fractional part is compensated by an intercalary 28th *nakṣatra* titled Abhijit. *Nakṣatra* computation appears to have been well known at the time of the *Ṛgveda*.

- Monojit Choudhury

Problem 5: Prussian Blue, Hungarian Red (10 marks)

Given below are some Hungarian[2] words and phrases, and their English translations, in random order.

fehérvérűség, hajhagyma, kék függöny, kékvér, piros kabát, piros rakennus, sötét vörös szem, sötét piros cipő, sötétkék szem, sötétzöld kupola, világoskék kabát, világos piros ruha, vöröshagyma, vörösvérsejt, zöldség

red blood cell, red coat, dark red shoes, dark red eyes, light red dress, red building, dark green dome, light blue coat, dark blue eyes, blue curtain, blue blood, vegetable, onion, hair bulb[3], leukemia[4]

Assignment 1: (6 marks)

Match the Hungarian words and phrases to their English translations.

Assignment 2: (1 marks)

Translate the following into English:

(a) kékeszöld

(b) fehérvérsejt

(c) hagyma kupola

Assignment 3: (3 marks)

Translate the following into Hungarian:

(a) red hair

(b) light red curtain

(c) dark blue building

(d) green onion

(e) dark red dome

– Monojit Choudhury

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[2]Hungarian is the official language of Hungary and an official language of the European Union. It is spoken by over 13 million people mostly in Hungary and neighboring countries.

[3]Hair bulbs are the source of hair in animals/humans, shaped like a tear drop or rain drop. They are located under the skin and continuously generate a single strand of hair.

[4]Leukemia is a type of cancer of the blood or bone marrow characterized by an abnormal increase of immature white blood cells called “blasts”.

Problem #5: Pashto (20 marks)

Given below are some Pashto sentences and their English translations:

1)	**Sangin kerkey maatawi**	Sangin is breaking the window.
2)	**Taaso waruna maatawel**	You were breaking the doors.
3)	**Sangin war maat ki**	Sangin has broken the door.
4)	**Taaso kerkey maata key**	You have broken the window.
5)	**Sangin war maatawi**	Sangin is breaking the door.
6)	**Sangin kerkeyi maataweli**	Sangin was breaking the windows.
7)	**Taaso waruna maatawey**	You are breaking the doors.
8)	**Taaso kerkeyi maati key**	You have broken the windows.
9)	**Sangin waruna maatawel**	Sangin was breaking the doors.

Translate the following English sentences to Pashto. Some sentences cannot be translated based just on the given data. If so, justify.

1) You have broken the door.
2) You have broken the doors.
3) Sangin has broken the windows.
4) Sangin is breaking the doors.
5) You are breaking the door.
6) Sangin was breaking the door.
7) You were breaking the windows.

Pashto is the native language of the *Pashtun* people of South-Central Asia belonging to the *Indo-Iranian language family*. Pashto is one of the two official languages of Afghanistan. It is also the second-largest regional language of Pakistan with an estimated 45–60 million speakers worldwide.

– Apoorva Bhagwat

Problem 5: Önge (20 marks)

Given below are some sentences in *Önge* along with their *corresponding* English translations.

Sl. No.	Önge	English
1	***mi chogegantitebe tukirikotalotaga***	*I pierced the fish with all the spears.*
2	***uemetabetebe***	*The dog drowned.*
3	***etia niko angibete belebe***	*We gave you a matchbox.*
4	***uemekotalotagatekkebe***	*All the dogs barked.*
5	***entoge ngialabukebe***	*The bachelor scolded you.*
6	***ueme ogentitebe***	*The dog pierced the husband.*
7	***onggege mi entueme belebe***	*The father gave me the puppy.*
8	***onggengegatekkebe***	*The old man barked.*
9	***allekotalota mialabukebe***	*All the women scolded me.*
10	***onggebe onggege choge belebe***	*The mother gave fish to the father.*
11	***etakotalota entuementitebe***	*All of us pierced the young dog.*
12	***entalle uema uebe chogega***	*The girl covered the dog with the fish.*

Assignment 1:

Translate the following sentences to *Önge*. If a sentence cannot be formed in *Önge* from the given data, write *"NP"* in your Answer Booklet and explain your reasoning.

a) *All the old women covered the matchbox with the spear.*
b) *The big dog barked.*
c) *The unmarried woman gave you the fish.*
d) *We pierced using all the spears.*
e) *You all drowned.*

Assignment 2:
Give words in *Önge* for the following:

a. *Old Woman*
b. *Man*
c. *Woman*
d. *Girl*
e. *Bachelor Man (yet to marry)*
f. *Boy*
g. *Mother*
h. *Father*

Önge is a language belonging to the *Andamanese-Ongan* language family, spoken by the Ongan people of South Andaman. It was once a widespread language in the Andaman Islands until the British colonizers arrived. *Önge* is on the verge of extinction: In 1997, it was estimated that there were only 96 native speakers of the language. The surviving speakers are located at Dugong Creek and South Bay in Little Andamans.

- Aalok Sathe

Problem #5: Making Sense of Turkish Words (20 marks)

Given below are some words in Turkish and their English translations, in no particular order.

gözlemci, döndürmek, gündöndü, gözlükcü, şarkici,
çocukluk, gözlemek, pazar, pazartesi, cumartesi, güneşli

English translations:

Saturday, Sunday, Monday, observer, singer
to observe, to rotate, sunny, sunflower, optician, childhood

Assignment 1:
Match the Turkish words with their English translations.

Assignment 2:
Translate the following words into Turkish:
observation, child, the state of being a singer, spectacles, Friday

Hint: Read all parts of the question carefully, before working out the solution.

Turkish belongs to the Turkic language family. It is spoken in Turkey and within smaller communities in many other countries. It is the mother tongue of more than 63 million people. ***g, d, p, t, c, s, z, m, k, l, ş, ç*** denote consonants, and ***ö, e, i, ü, a*** are vowels.

– Monojit Choudhury

Problem #1: Lepcha (15 marks)

Sikkim has 11 official languages, which are listed below in English as well as the *Lepcha Script*.

Nepali ᰟᰛᰡᰣ, **Lepcha** ᰡᰮᰜ, **Sikkimese** ᰠᰮᰜᰧᰮ, **Tamang** ᰚᰜᰛᰜ, **Limbu** ᰡᰮᰣᰮ, **Newar** ᰟᰮᰰᰜ, **Rai** ᰮᰡᰧ, **Gurung** ᰜᰮᰜᰮ, **Magar** ᰛᰰᰜ, **Sunwar** ᰤᰮᰰᰜ, **English** ᰜᰧᰡᰜᰮᰚ

Assignment 1:

The *Lepcha* name for one of the above languages does not match its English name. Which one? It is actually written as **Drendzongke** (*dz* is pronounced somewhat like the *j* in jug.)

Assignment 2:

The *Lepcha* speakers, who call themselves the **Roong haagiit** (ᰜᰮᰧ ᰬᰡᰰᰜ), are composed of four main distinct communities: ᰧᰮᰠᰛ, ᰚᰜᰤᰜᰛ, ᰡᰬᰰᰜᰛ, ᰡᰮᰛ

Transcribe these four community names into English.

Assignment 3:

Sikkim boasts of the third highest mountain peak of the world, *the Kangchenjunga*. Transcribe this name in the Lepcha script, considering the fact that it is pronounced as **Kaang-chen-dzong-gaa** (means "*the five treasures of the high snow*" in the Tibetan language).

The *Lepcha* script, used for writing the *Lepcha* language, is derived from the *Tibetan* script, and may have some Burmese influence. Around 30 to 50 thousand people speak the *Lepcha* language, who are spread across Sikkim, Northern West Bengal, Bhutan and Nepal.

– Monojit Choudhury

3. Kurmali (20 marks)

Problem designed by Bornini Lahiri

Here are sentences in Kurmali and their respective English translations.

1.	*moi bat khau*	*I ate rice*
2.	*moira bat khayonu*	*We ate rice*
3.	*toi choani ke dekhliae*	*You (Masculine) saw a girl*
4.	*toira choara ke dekhlini*	*You all saw the boys.*
5.	*choara choani ke dekhaone*	*Boys saw a girl.*
6.	*choara ghar dekhaonu*	*Boys saw the house.*
7.	*kokur kha*	*Dog ate.*
8.	*kokura khayon*	*Dogs ate.*
9.	*toi kokur ke dekhleou*	*You (feminine) saw a dog.*
10.	*Radha Salman ke dekhuoi*	*Radha saw Salman.*
11.	*choanira bhat khaenu*	*Girls ate rice.*
12.	*Salman Radha ke putuli deaoe*	*Salman gave a doll to Radha.*

Assignment 1 [Answer Booklet]**:** Translate the following sentences in Kurmali.

1. I saw Radha.
2. I gave Salman rice.
3. Girls ate the rice.
4. A girl gave Radha dolls.
5. Radha saw the boys.
6. We saw a dog.

Assignment 2: Explain how the grammar of the language works.

Notes: (1) Kurmali is a language spoken in some of the districts of West Bengal, Jharkhand, Orissa and Madhya Pradesh. It is believed to be a mixed language. (2) The data has been modified slightly for ease of solvability.

Problem 4: Kunimaipa (15 marks)

Sohopuho noho.	I went and then ate.
Sata noho.	I went and ate (same time).
Seŋipuho neŋi.	You went and then ate.
Sahapuho naha.	He went and then ate.
Hehopuho soho.	I stayed and then went.
Hehana soho.	He stayed and then I went.
Sahana noho.	He went and then I ate.
Sahana neŋi.	He went and then you ate.
Hemapuho soma.	I will stay and then go.
Hekepuho seke.	You will stay and then go.
Hepanepuho sapane.	He will stay and then go.
Heta naha.	He will stay and eat (same time).
Hemana seke.	I will stay and then you will go.
Hekena soma.	You will stay and then I will go.
Hepanena seke.	He will stay and then you will go.

Assignment 1: (5 marks)

Translate the following into English:

(a) Hehapuho saha

(b) Nata hema

(c) Heŋina saha

Assignment 2: (5 marks)

Translate the following into Kunimaipa:

(a) I will eat and then go.

(b) You will eat and go (at the same time).

(c) I will stay and then he will eat.

Assignment 3: (5 marks)

How are words or parts of words put together to form a valid Kunimaipa sentence? What are their meanings and functions?

– Zarana Sarda

Kubeo

Given below is a list of Kubeo[1] words with their translations, in **no particular order**. Note that two Kubeo words given translate to 'cave' in English.

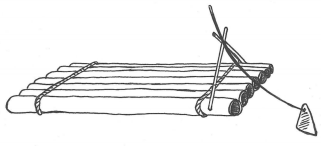
You are given the fact that jabe means 'seed' in Kubeo. You are also given some pictures that illustrate what some of these words mean. Using this information, match the words in the lists given below.

(a) dẽĩjabe (b) dẽĩkũ (c) xiajokũ (d) tãũjo (e) tãũbɨ (f) hokebɨ (g) ũejo (h) ũekobe (i) kɨ̃ɨ̃ãkobe (j) kɨɨ̃ãjãbĩ (k) kɨɨ̃ãbo (l) jãbĩ (m) xia

(1) river (2) house (3) cave (4) cave (5) rat's nose (6) heap of moriche fruit (7) fishtrap (8) nostril (9) raft (10) iron nail (11) rock (12) metal can (13) moriche seed

Note: Moriche palm is a palm tree species that grows in and near swamps and other wet areas in tropical South America.

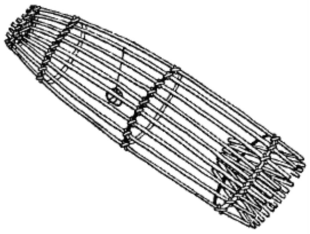
A raft is a means of water transport which has traditionally been constructed of wood or reeds tied together to stay afloat a water body.



Raft



Rat



Fishtrap

Assignment 1:

Here is a list of some of the units of meaning that make up these words. Guess what they mean on their own.

(1) dẽĩ (2) kũ (3) jo (4) tãũ (5) bɨ (6) ũe (7) hoke (8) kobe (9) kɨrɨ̃

[1]Kubeo is a language from the Eastern Tukanoan family, spoken in the Northwest Amazon.

Assignment 2:

Here are the meanings of some additional words in Kubeo:

Kubeo	English
toiwai	writing
bẽ	thread
bú͡tʃi	tobacco
hɨoi	healing
jukira	salt
toa	fire, light

Using these, translate the following Diaboa words to English:

(1) toiwaixũe (2) tãũbẽ (3) hɨoijãbĩ (4) jukirajabe (5) toabɨ

– Abhinav Menon and Mukund Choudhary

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.

Kholosi

The following is a scrambled list of Kholosi[1] sentences and their English translations.

1. ɔnguloɔn
2. hiki sokoli gazzi
3. mɔy hiki motor ke pendemay
4. ho tɔffen ke cʰenemay
5. vyɔw ɔngulos te gazzien ke cʰenai
6. tiriṛi śurkiu tiriṛi motor ke pendavɔn
7. Nilufar segaw tɔffen ke pesaiya
8. tereḷɔ javɔn vyɔwkɔ Ali ke barɔ ketɔb genavɔn
9. mɔy gohroy kɔro e
10. śurki motoren ke pendai
11. Nilufar gohros sokolo e

(a) Nilufar is able to see apples.

(b) I am looking at one car.

(c) My horse is black.

(d) The three young boys gave Ali two books.

(e) The girl saw cars.

(f) one white rope

(g) The boy cut the ropes with his finger.

(h) Nilufar’s horse is white.

(i) He is cutting the apples.

(j) Three girls saw three cars.

(k) their finger

Assignment 1:

Match the Kholosi sentences with their correct English translations.

[1]Kholosi is an Indo-Aryan language spoken in two villages (Kholus and Gotav) in Hormozgan, Iran.

Assignment 2:

Translate into Kholosi.

(a) their horse

(b) Three girls cut the ropes.

(c) Three boys cut the ropes.

(d) Ali's apple is white.

(e) He cut the apples.

(f) The girl saw the rope.

(g) The girls saw the apples.

(h) The girls gave the rope to Nilufar.

– Aryaman Arora

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.

Problem #5: Japanese Compound Words (20 marks)

In Japanese, certain root words undergo a change at their left edge when they form compound words under specific circumstances. This phenomenon is traditionally called *Rendaku*. Examine the following examples of Japanese compound words:

Japanese words	Compound word	Gloss	Meaning
hito* + *hito	***hitobito***	person + person	people
ike* + *hana	***ikebana***	keep alive + flower	flower arrangement
toki* + *toki	***tokidoki***	time + time	sometimes
te* + *kami	***tegami***	hand + paper	letter
hitori* + *tabi	***hitoritabi***	one person + travel	travelling alone
tsuno* + *tokage	***tsunotokage***	horn + lizard	horned lizard
yama* + *kawa	***yamakawa***	mountains + rivers	mountains and rivers
aka* + *huda	***akahuda***	red + tag	red tag
maki* + *susi	***makizusi***	rolling + sushi#	rolled sushi
yama* + *kawa	***yamagawa***	mountain + river	mountain's river
hebi* + *tosi	***hebidosi***	snake + year	snake year
eda* + *ha	***edaha***	branches + leaves	branches and leaves
oo* + *kaze	***ookaze***	big + wind	big wind
kokoro* + *tsukai	***kokorodzukai***	heart + using	anxiety
denki* + *kurage	***denkikurage***	electric + jellyfish	electric jellyfish
yuki* + *kaeri	***yukikaeri***	coming + going	coming and going
hana* + *sono	***hanazono***	flower + garden	flower garden
nise* + *karasu	***nisegarasu***	fake + crow	fake crow

Assignment 1:
Fill in the following compound words.

Japanese words	Compound word	Gloss	Meaning
wa* + *kasi	_____	Japanese + cake	Japanese cake
yama* + *kazi	_____	mountain + fire	forest fire
juzu* + *tsunagi	_____	rosary + sequence	roping together
yama* + *tera	_____	mountain + temple	mountain temple
ao* + *kawazu	_____	green + frog	green frog
uwa* + *huta	_____	top + lid	top lid
e* + *tako	_____	picture + kite	picture kite

Assignment 2:
When compounds are made up of more than two words, things become truly interesting. Fill in the blank entries (A to E) in the following Table:

Japanese words	Compound Word	Gloss	Meaning
hosi-kaki-tsukuri	***hosi-gaki-dzukuri***	dried persimmon* making	making dried persimmon
nise-kami-tana	***nise-kami-dana***	fake divine altar	a divine altar that is fake
nise-kami-tana	***nise-gami-dana***	fake divine altar	an altar for a fake God
ori-kami-tana	***ori-gami-tana***	fold paper shelf	origami shelf
nise-sakura-dayori	***A. ?***	fake cherry-blossom tidings	tidings of fake cherry-blossoms
nise-sakura-dayori	***nise-sakura-dayori***	fake cherry-blossom tidings	fake tidings of

			cherry-blossoms
nuri-hasi-hako	***B. ?***	lacquered chopstick box	a box for lacquered chopsticks
nuri-hasi-hako	***nuri-hasi-bako***	lacquered chopstick box	a lacquered box for chopsticks
nise-tanuki-siru	***C. ?***	pseudo tanuki** soup	soup made from pseudo tanuki
nise-tanuki-siru	***D. ?***	pseudo tanuki soup	tanuki soup that is not authentic
oya-ko-kenka	***E. ?***	parent child quarrel	parent and child quarrel

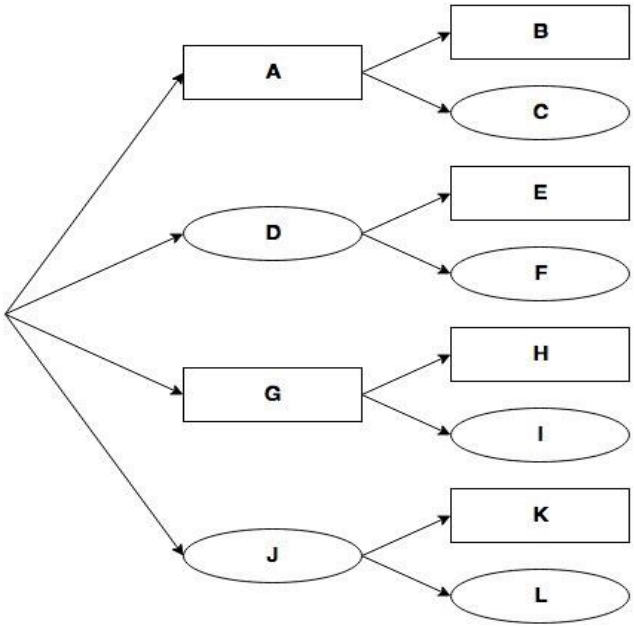
Japanese is spoken by about 125 million speakers primarily in Japan.
#**sushi** is cooked vinegared rice combined usually with raw fish.
***persimmon** is a fruit.
****tanuki** is Japanese raccoon dog.

- *Zarana Sarda*

1. Imya Imyevna Rossiyov

Problem designed by Saujas Vaduguru

You are studying a Russian language and culture course in college, and as part of an assignment, you have to figure out how the Russian family naming system works. You are given the scheme of a family tree, and the statements by people where they name some people to whom they are related. These relatives may be their parent, child, or sibling. The statements include names of some people who do not fit in the given scheme for the family tree. In the figure, an arrow goes from a parent to a child. Rounded nodes are for female names and rectangular ones are for male names.



The following statements (naming relatives) are from members represented in the family tree — A through L. But, the statements themselves may contain names of people who are not represented in the family tree, i.e., the statements may name people who are not one of A-L.

Note that (1) *Vladimir* is male, and *Yekaterina* is female. (2) There are two different *Olga*s in the family tree. (3) The first names are followed by corresponding nicknames in brackets.

Vladimir (Vova): *Vanyechka, Andriy Vladimirovich Ivanov, Olga Vladimirovna Ivanova*
Viktor (Vitya): *Alexandr Romanovich Smirnov, Natalya*
Valeriya (Lera): *Anna Vladimirovna Ivanova, Sergey Pavelovich Popov*
Anna (Anya): *Lyonyechka, Lerochka, Olga Vladimirovna Ivanova, Andriy Vladimirovich Ivanov*
Olga (Olya): *Vityechka, Natashochka, Anna Vladimirovna Ivanova, Andriy Vladimirovich Ivanov, Vladimir Vladmirovich Ivanov*
Andriy: *Vladimir, Olechka, Boryechka*
Leonid (Lyonya): *Anna Vladimirovna Ivanova, Valeriya Sergeyevna Popova*
Nataliya (Natalya): *Viktor Alexandrovich Smirnov*
Olga (Olya): *Boris, Andriy Vladimirovich Ivanov*
Ivan (Vanya): *Yelena, Yekaterina Andriyevna Mikhailova*
Boris (Borya): *Andriy Vladimirovich Ivanov, Olga Andriyevna Ivanova, Alexandra Georgyevna Vasileva*
Yelena (Lena): *Yekaterina Andriyevna Mikhailova, Vladimir Vladmirovich Ivanov*

Assignment 1 [Answer Booklet]: Fill in the full, three-part names of the family members A-L in the family tree. If there are multiple possible solutions, write only one.

Assignment 2: Explain how the Russian naming system works.

Assignment 3 [Answer Booklet]**:** There is a grammatical error in the title of the problem. What is it?

Notes: Russian is an East Slavic language and an official language in Russia, Belarus, Kazakhstan, Kyrgyzstan and many minor or unrecognized territories. It is the most geographically widespread language of Eurasia. It is also the largest native language in Europe, with 144 million native speakers in Russia, Ukraine and Belarus.

Problem no 1. Sudanese Arabic

Examine the data from Sudanese Arabic given below and answer the questions below. <u>Set A</u>

i a. sirib	'he drank'	i b. hamal	'he carried'
ii a. siribta	'I drank'	ii b. hamalta	'I carried'
iii a. siribti	'you fem. drank'	iii b. hamalti	'you fem. carried'
iv a. siribtu	'you pl. drank'	iv b. hamaltu	'you pl. carried'
v a. siribna	'we drank'	v b. hamalna	'we carried'

Identify the past form of the verb stems in Set A and isolate the 3rd sg.mas., 1st sg, 2nd fem, 2 pl and 1st pl markers in the data above.

<u>Set B</u>

i a. sirbat	'she drank'	ii b. hamalat	'she carried'
ii a. sirbu	'they drank'	ii b. hamalu	'they carried'

Isolate the 3rd fem.sg. and 3rd pl. markers.

Account for the alternations in the stem meaning 'drank', contrasting it with the lack of alternation in the stem meaning 'carried'.

<u>Set C</u>

i a. hamalti	'you fem.carried'	i b. hamaltii	'you fem carried him'
i c. hamaltiina	'you fem.carried us'	i d. hamaltiiha	'you fem.carried her'
i e. hamaltiihum	'you fem.carried them'		

Isolate the object markers and specify the order of attachment of the subject and object markers.

Predict the words for 'we carried him' and 'you pl carried her'?

Problem No. 1. [8 Marks]

On the basis of the data provided, answer the question(s) that follow:

Hebrew

diber 'he spoke'
hasav 'he thought'
sagra 'she shut'
hasva 'she thought'
kalat 'he received'
kalta 'she received'
milmel 'he muttered'
kimet 'he wrinkled'

Translate the following -
dibra
kimta
milmla
sagar

Problem No. 6. [20 Marks]

A. Examine the data of an English child and answer the following questions: (10 Marks)

1. Don't giggle me [=make me giggle].
2. He cried me [=made me cry].
3. But I can't eat her. [= feed]
4. Please see me the book. [=show]
5. I disappeared the bear at the back of the car. [=made the bear disappear]
6. Who died him? [=killed]

Identify the pattern of errors in the child language data. Attempt an explanation for this pattern.

B. Examine the data from child language given below: [10 Marks]

<u>Set A</u>

i.	[bek]	'peg'	ii.	[gi]	'key'
iii.	[gʌm]	'come'	iv.	[di]	'tea'
v.	[bæk]	'bag'	vi.	[bæt]	'bad'
vii.	[bæt]	'pad'	viii.	[bæt]	'bat'

<u>Set B</u>

i.	[dut]	'juice'	ii.	[don]	'John'
iii.	[deidi]	'lady'	iv.	[dait]	'light'
v.	[deidi]	'lazy'			

a. **Account for the changes in consonants in the child's pronunciation (informally). How many sounds in the adult vocabulary are realized as [d] in the child's pronunciation?**
b. **Can you predict how the child is likely to pronounce the word 'Sally'?**

(c) dark blue building

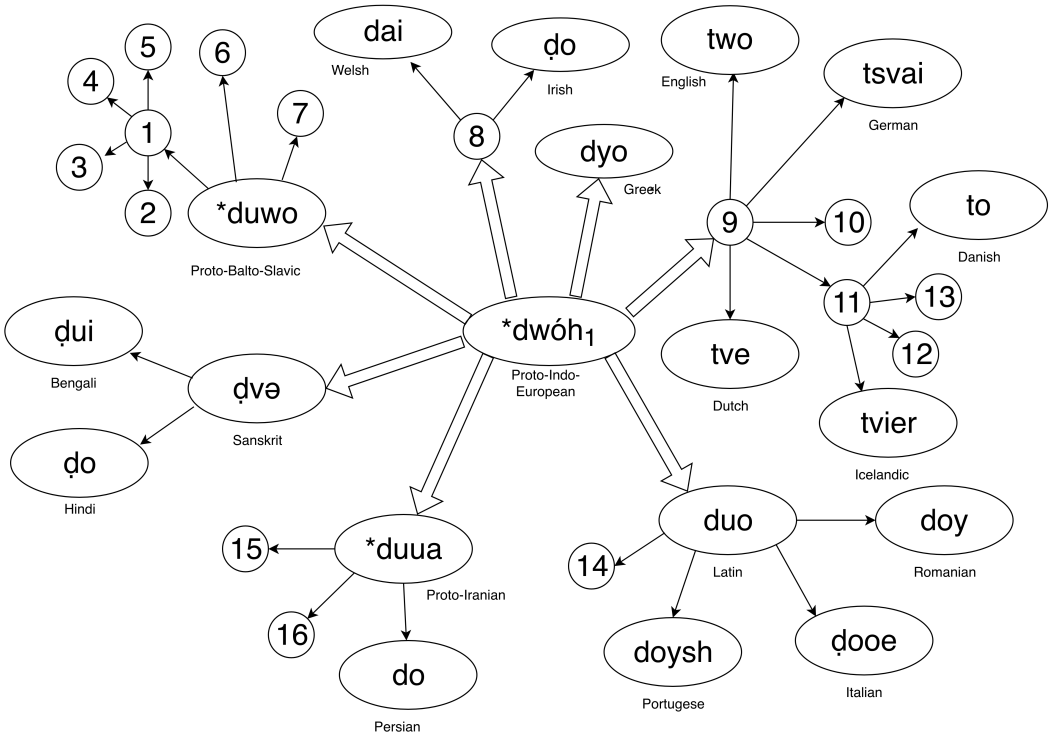
(d) green onion

(e) dark red dome

– Monojit Choudhury

Problem 4: From Two to Two, Too (14 marks)

One way that languages develop is through the evolution and gradual change of existing languages. Because of this process, like many other words, the word for the number 2 ("two") is similar in many related languages. Research shows that the evolution chart may look like this (some words are missing):



In the chart, all words are spelled according to how they are pronounced. For example, do in Persian rhymes with 'show', unlike the English word 'do'. The only exception to this is the English word 'two' which is spelled as normal but it is not pronounced with a 'w' sound. (Fun fact: around 1,000 years ago, the 'w' sound was actually present in English!) The character 'ḍ' indicates a 'd' sound produced near the teeth rather than the roof of the mouth, and the character 'ə' is the sound when you say 'uhhh'.

Some languages have 'proto-' before them and the words are marked with *. This is because they are hypothetical languages that linguists have tried to reconstruct from historial evidence, and we are not sure about the words or the pronunciations; 'proto-'

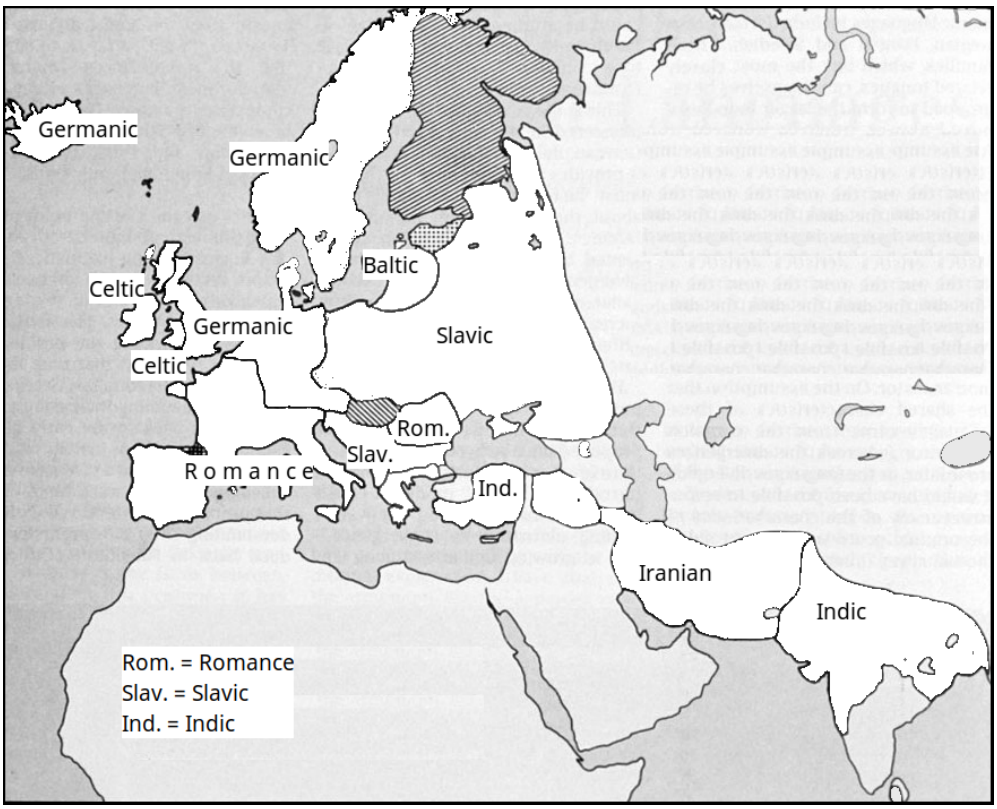
means something that came before, or the original form of something. 'Slavic', 'Baltic', 'Celtic', 'Germanic', 'Iranian', 'Indic', etc. are names of large language families.

Assignment 1: (14 marks)

Fill in the missing words numbered (1)–(16) in the chart above from the following list:

Word	Language	Word	Language
dva	Serbo-Croatian	*dwaw	Proto-Celtic
dva	Polish	tsvey	Yiddish
dva	Czech	*dva	Proto-Slavic
too	Norwegian	du	Lithuanian
dva	Russian	divi	Latvian
*twai	Proto-Germanic	do	Balochi
ḍos	Spanish	du	Kurdish
tveir	Old Norse	tvo	Swedish

Hint: Kurdish and Balochi have evolved from the same language, and Lithuanian and Latvian are Baltic languages. Also, this map of language families may be useful.



– Mukund Choudhary and Shardul Chiplunkar

Problem #6: Finnish (20 marks)

Following are some words in English and their Finnish translations, in no particular order:

English

cellphone, to talk, time, e-cash, time traveler, to write, airplane, blog, to type, telephone, computer, to hatch, Information Technology, bus, service, e-mail, incubator

Finnish

hautoma, aikamatkailija, sähköposti, linja, hautomakone, matkapuhelin, nettipäiväkirja, sähköinen raha, kirjoitus, tietotekniikka, lentokone, sähkönkirjoitus, aika, tietokone, puhelin, puhua, linja-auto

Assignment 1
Figure out the correct correspondences.

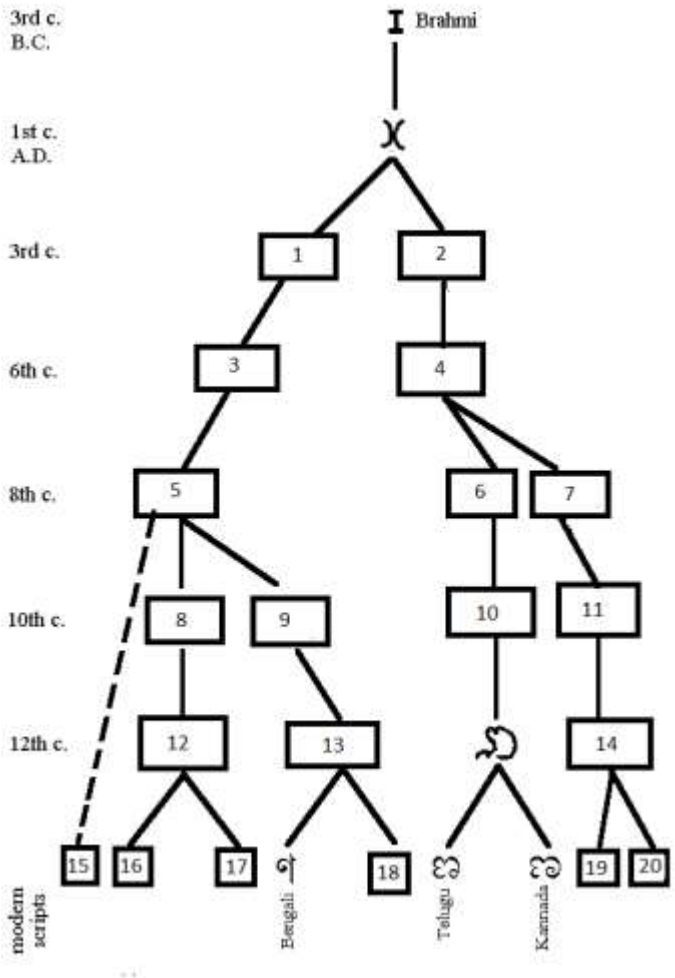
Assignment 2
Translate to Finnish:
Diary, Journey, Flight / To Fly

Assignment 3
Translate to English: **matkatietokone, sähkön, tieto linja, kirjoituskone**

Finnish is spoken by about five million people, most of whom reside in Finland. It is a member of the *Finnic group* of the *Uralic family* of languages. The Finnic group also includes *Estonian* and a few minority languages spoken around the Baltic Sea.
a, e, i, o, u, ä, ö and **y** are vowels and rest of the letters are consonants.

- Rajan Dalal

Problem #1: Evolution of *Nna* (15 marks)



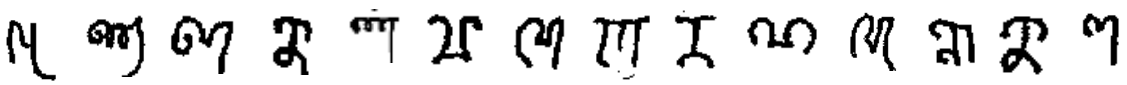
The phylogenetic tree depicting the evolution of the glyphs for the grapheme *Nna* (retroflex nasal ɳ)

All the modern Indian language scripts have evolved from an ancient script called *Brahmi*, which was in use around 3rd century BC. Observe the forms (*glyphs*) of the letter (*grapheme*) *Nna* in 9 modern Indian scripts: Devanagari: ण , Bengali: ণ , Gujarati: ણ, Gurumukhi (Punjabi): ਣ , Kannada: ಣ, Malayalam: ണ Odiya: ଣ , Tamil: ண Telugu: ణ.

Linguists believe that all these forms have developed from the Brahmi glyph 𑀡. Based on the several intermediate glyphs of this grapheme that has been found in the historical documents, linguists have been able to create the evolutionary history of the grapheme *Nna*, represented in the form of a *phylogenetic tree* or *evolutionary tree* as shown in the left. In this tree, bottom-most boxes denote the glyphs in the modern scripts and higher boxes denote older forms from which the later glyphs have evolved. (Most of the intermediate glyphs have been omitted and replaced by numbers.) For instance, glyph #9 (used in 10th century) has given rise to glyph #13 (used in 12th century) which in turn has led to the development of the glyphs used in modern Bengali and

Assignment 1

Your task is to complete the above phylogenetic tree by filling the boxes with the appropriate glyphs given below or the modern forms written above.



A B C D E F G H I J K L M N

Write down your answer for each box by specifying the letter (A-N) of the glyph that goes in the box. For modern forms, use the language/script name.

Assignment 2

Modern scripts and the glyphs have remained almost fixed for over 300 years now, and are unlikely to change in the future. Invention of which revolutionary technology in the 15th century has led to this situation? Explain how this technology has prevented in further evolution of the structure of letters.

Brāhmī is one of the oldest writing systems used in the Indian subcontinent and in Central Asia during the final centuries BCE and the early centuries CE. It is the ancestor of all the modern Indic scripts as well as the Thai, Lao, Tibetan and Burmese scripts. In modern times, it was deciphered by James Prinsep, a British archaeologist, in 1837.

Nna is the Unicode name for the letter representing the *nasal retroflex* /ɳ/ – a nasal sound produced by curling the tongue. Retroflex sounds, such as /ɳ/, /ʈ/ and /ɖ/ are found in several Indic and South Asian languages.

- *Monojit Choudhury*

Problem 7: Evoke Kove (20 marks)

Given below are some sentences in Kove[5] and their English translations.

yau ngaoli vula pa tuanga	I bought a shell necklace from the village.
veao ukonaghau	You saw me.
Neti ikonari vula noha	Neti saw the shell necklaces yesterday.
Neti ipasolani tuanga pari kekele	Neti showed the children the villages.
yau ngaaniri niu pa tuanga	I ate coconuts in the village.
tamone tipasolaniri vula pa ghau	The men showed me the shell necklaces.
veao upasolani niu pari kaua	You showed the dogs the coconuts.
Neti ihaugho	Neti hit you.
ghaya iani niu lima	The pig ate five coconuts.
yau ngaoli kaua pari tamone	I bought the dogs from the men.
tamone tikona vula tolu	The men saw three shell necklaces.
veai ikonari tamone pa tuanga	He/she saw the men in the villages.

Assignment 1: (8 marks)

Translate the following sentences into Kove.

(a) Neti ate a coconut.

(b) I showed you five shell necklaces.

(c) The pig bought coconuts from the children.

(d) I saw the villages yesterday.

Assignment 2: (12 marks)

How are words or parts of words put together to form a valid Kove sentence? What are their meanings and functions? (Try to make your explanation fit all of the data. None of the given sentences should be an exception to your answer.)

– Rujul Gandhi

This is the end.
We would like to hear your feedback about this exam!
After the exam is over, please visit `tiny.cc/plofeedback2019-senior`.

[5] Kove is an Austronesian language spoken by about 6,750 people in the West New Britain province of Papua New Guinea.

2. Ethiopian Months (20 marks)

Problem designed by Monojit Choudhury

Here are the names of the Ethiopian months in the *Fidel* script along with their beginning dates:

መስከረም	ጥቅምት	ሕዳር	ታሕሣሥ	ጥር	የካቲት	መጋቢት
11 or 12 Sep	11 or 12 Oct	10 or 11 Nov	10 or 11 Dec	9 or 10 Jan	8 or 9 Feb	10 Mar
ሚያዝያ	ግንቦት	ሰኔ	ሐምሌ	ነሐሴ	ጳጉሜ	
9 Apr	9 May	8 Jun	8 Jul	7 Aug	6 Sep	

The transcriptions of the month names, in no particular order, are:

Ṗagume, Gənbot, Taḫśaś, Ḥamle, Ḫədar, Mägabit, Ṭəqəmt,
Yäkatit, Mäskäräm, Nähase, Säne, Ṭərr, Miyazya

Assignment 1 [Answer Booklet]: Match the names of months written in the Fidel script with their transcriptions.

Assignment 2 [Answer Booklet]: Sunday, እሑድ and Monday, ሳኞ are called *əhud* and *sañño* respectively, in Amharic. Transcribe the names of the other days using the same transcription scheme

ማክሰኞ (Tuesday), ረቡ (Wednesday), አርብ (Friday), ሐሙስ (Thursday), ቅዳሜ (Saturday)

Assignment 3 [Answer Booklet]: Write down in Fidel script the Amharic word for "Ethiopia", which is transcribed as *ityoṗṗya* and contains the letter ዮ .

Notes: (1) Amharic is a Semitic language spoken in Ethiopia. It is the native tongue of more than 25 million people and is the second most spoken Semitic language after Arabic. (2) The consonants are pronounced like in English; a dot below the consonant (*Ḥ* and *Ṭ*) means that it is an *ejective* (pronounced with a kind of stress); *q* is the ejective form of *k*; *ś* is pronounced like *ch* in *church*; *ä* and *ə* are vowels.

Problem 1: Estonian Word Forms (15 marks)

Given below are some groups of words in Estonian[1] and their corresponding English translations. Each row contains words from the same group.

jooksmine	the action of running	jooksma	to run	jookslema	to run about
pikk	long	pikkus	length	pikkenema	to lengthen
lendus	flight	lendama	(1)	(A)	to flutter
(B)	to study	(C)	student	õpimine	(2)
suur	(3)	(D)	greatness	suurenema	to enlarge
voolav	fluid	(E)	fluidity	(F)	the action of flowing
välkatama	to flash once	välkama	(4)	välklema	to flicker
(G)	to know	(H)	science	teadlane	scientist
kangema	(5)	kangus	strength	(I)	hero
jumalasu	god/goddess	jumalasus	(6)	jumalasulane	priest
toukama	(7)	touklema	to jostle	toukamine	the action of pushing
selge	(8)	(J)	clarity	selgenema	(9)
(K)	abstinence	karsklane	abstainer	(L)	the action of abstaining
peen	(10)	(M)	finesse	peenenema	to refine
klõpsatama	to click once	klõpsma	to click	klõps	(11)
praksatama	to crack once	praksuma	(12)		
rändama	(13)	(N)	to wander		
(O)	honey	(P)	honeybee		
paks	thick	(Q)	to thicken		

Assignment 1: (12 marks)

Fill in the blank cells (A-Q) with Estonian words and the blank cells (1-13) with English words.

Assignment 2: (3 marks)

Given an English phrase, how would you form the corresponding Estonian word?

– Antara Raaghavi Bhattacharya and Aparna Gupte

[1]The Estonian language is a Uralic language of the Finnic branch spoken in Estonia. It is the official language of Estonia, spoken natively by about 1.1 million people.

Esimbi

Here are two sequences of five numerals written in Esimbi[1]. In each sequence, the ratio between each number and the next is the same. Note that this ratio may or may not be different across the two sequences.

Sequence 1

(a) mərakpə

(b) moŋi

(c) moŋioŋi

(d) buɣu nə malalə

(e) mobuɣu makələ nə mərakpə

Sequence 2

(a) kenə

(b) makələ

(c) matənə moŋi

(d) mobuɣu mɔrakpə nə matənə mərakpə

(e) mobuɣu moŋioŋi nə kenə

Assignment 1:

Find the Hindu-Arabic numerals (i.e. the numbers you usually write) that correspond to the two sequences above.

Assignment 2:

Following are three expressions in Esimbi, where the blank boxes are placeholders for the missing operators. All three expressions evaluate to the same number, 42.

(a) mobuɣu buɣu nə mobuɣu mɔrakpə nə malalə ☐ makələ

(b) mobuɣu matənə nə kenə ☐ matənə moŋi

(c) mobuɣu mɔrakpə nə kenə ☐ mərakpə

Write 42 in Esimbi numerals. Fill in the missing operators and write the equations in Hindu-Arabic numerals.

– Mukund Choudhary and Aparna Gupte

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.

[1] Esimbi is a Tivoid language of southwestern Cameroon. It is also called Isimbi, Simpi, Age, Aage, Bogue and Mburugam.

Problem 6: Epic Greek Poetry (20 marks)

The Ancient Greeks were well-known for their love of mathematics, sciences, the arts and, of course, poetry. The great epics, the "Iliad" and the "Odyssey" were composed by Homer around 3000 years ago.

As with most Greek Poetry, Homer's poems followed a *rhythmic meter*, so that each line contained syllables in a well-defined pattern. Homer's epics, in particular, followed the *Dactylic Hexameter*, where each line of the poem contains 6 metrical 'feet'. The first 5 *'feet'* are identical, but the 6th *'foot'* is shorter than the rest.

For this problem, the translations of the words are not important. Tone is also not important. This problem requires you to analyze rhythmic meters and find a general pattern in the arrangement of syllables.

Each of the following verses has been adapted from Homer's "Odyssey" and follows the *Dactylic Hexameter*:

ândra moi ênnepe, Moûsa, po lûtropon, hôs mala pôlla
plângthe, e peî Troli nês hiler ôn ptoli êthron e pêrse;
pôllone d'ânthrope pôn iden âstela kaî nolon êgno,
pôlla d'ho g'ên nopon tô pathen âlgela hôn kata thûmon
ârnumen ôs luthen tê hipsuch ên makai nôston he taîron.
âll' maloud' hôs heta roûs kaper rûsato, îemen ôs per;
aûtomon gâr spheter êsin a tâsthali eîsin o lônto,
nêpiloi, hoî kata boûs hipe îonos êeli oîo
êsthilon; aûtar ho toîsin a pheîleto nôstimon êmar.
tôn hamo thênge, the â, thuga têr Dios, eîpe kai hêmin.
Zênos en î mega roîsin Olýmpia hâthrokoi êsan.
Laêrten he rôa, ton oûketi phâsi po lînde
Pâllamas Âthena îe, ku rê Dikos, ôphra min aûton
ôuthʾa Hêrakel î okutʾ Eûrytoi Oîchali êi,
ôichetʾ, Ôdysse ôs mera gâthymou phaîdimon hyîon

To help you understand the syllable-structure of Homer's Epics, here are ten more verses in Ancient Greek. These verses could not have been part of the "Odyssey" because they do not follow the *Dactylic Hexameter*.

labôntes de, ân dê kaì autoî
amphoterô: Pelise men en eurychoroi Iaolkoi
autar ho gymnothe rhakeon polymetis Odysseus,
Hêliyos d'anôro ûse, li pôn peri kallêla limnên,
âîpsâ dʾê mîos hêta rôisin pot rŷnas ê kêlsâ
kéklyte nŷn, Ithakḗsioi, hótti:
lygrón, hòn ek Trôlîes êpêteilâtô Pallâs Âthêne.
Zeûs pater alloi mâkares theoî aiên eôntes,
égreo, Pênelopêia, phîlon ku tekôs, ôphra min idêai
hôs êphamen, hô êpeita Pôseidoni ânakti

Assignment:
Here are five more verses in Ancient Greek. Which of these verses could have been part of Homer's Epics? In other words, which verses follow the *Dactylic Hexameter*?
If a verse follows the *Dactylic Hexameter*, mark *"Y"* next to the corresponding verse number in your Answer Booklet. Otherwise, mark *"N"*.

a) ***ângeliên periphrô Pênelopêia,***

b) ***Têlemachʾ, oûketi kâla domôn apo têkʾ alalēsai,***

c) ***hôs ephathʾ Hêrmei pot, âllʾ ophra Aîgisthoi hâthrokoi hêmin***

d) ***tôn dʾemeîbetʾ epeîta Gerêniôs hippôta Nêstor:***

e) ***oûte pôtʾ ein agora dîchʾ êbazomên oûtʾ eni boûle,***

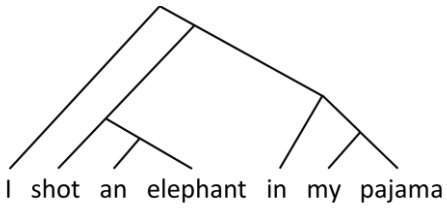
Be sure to describe the structure of the *Dactylic Hexameter* in Homer's Epics in your explanation.

1. A syllable is a unit of pronunciation uttered without interruption, loosely, a single sound. All words are made from at least one syllable. **2.** A 'foot' is an arbitrary unit in poetry consisting of a definite pattern of syllables. **3.** â, ê, î, ô, û and ŷ are specific vowels in Ancient Greek. hʾ, tʾ, d', g' and l' are specific consonants in Ancient Greek.

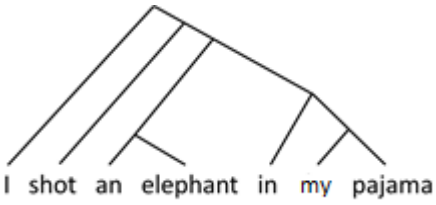
- Sesh Sadasivam

Problem #6: Encoding Trees (20 marks)

Linguistic structures, such as grammatical relations between words in a sentence, are often represented as a *Tree*. For example, the following *parse* or *phrase-structure trees* represent two different interpretations of the same sentence.



Tree-*A*



Tree-*B*

The above trees are known as strictly *binary trees* because at every branching point of the tree, there are *exactly two* branches – a left and a right one. Since drawing a tree takes more space and time, trees are often encoded as strings. According to one such encoding scheme for binary trees, the above trees are represented as:

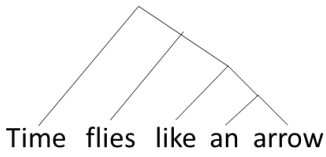
Tree-*A*: *I* 3 *shot* 1 *an* 0 *elephant* 2 *in* 1 *my* 0 *pajama*

Tree-*B*: *I* 4 *shot* 3 *an* 0 *elephant* 2 *in* 1 *my* 0 *pajama*

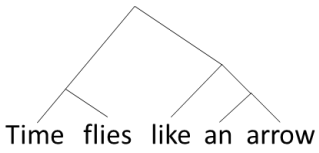
For this encoding scheme, every distinct binary tree has a unique string representation and for every distinct *valid* string representation (like the two examples above), one can also uniquely construct a binary tree. Your task is to figure out this encoding scheme and prove certain facts about it.

Assignment 1:

Shown below are two phrase-structure trees, **Tree-*C*** and **Tree-*D*** for two different interpretations of a sentence. Write down the string encodings for **Tree-*C*** and **Tree-*D*.** Explain your assumptions.



Tree-*C*



Tree-*D*

Assignment 2:

Construct the trees corresponding to the following encoded representations (words have been replaced by letters): **a 0 b 3 c 0 d 1 e 2 f 0 g** **a 0 b 1 c 2 d 3 e 4 f**

Assignment 3:

The following encodings do not represent binary trees. Convert them into valid tree encodings by incrementing or decrementing only one of the numbers by exactly 1.

a 0 b 2 c 0 d 2 e 1 f 0 g **a 1 b 3 c 2 d 1 e 0 f**

There are several constraints that a string must satisfy to be a valid encoding for a binary tree. What constraints do these strings violate, which makes them invalid?

Assignment 4:

Let $a_1\ x_1\ a_2\ x_2\ a_3\ \ldots\ a_{n-1}\ x_{n-1}\ a_n$ be a valid representation of a binary tree for a n word sentence $a_1\ a_2\ a_3\ \ldots\ a_{n-1}\ a_n$ where $x_1, x_2, \ldots x_{n-1}$ are the numbers separating the words. Prove that

$$2^n - 1 \geq \sum_{i=1}^{n-1} 2^{x_i} \geq \lfloor \log_2 n \rfloor \left\lfloor \frac{n}{2} \right\rfloor$$

- *Monojit Choudhury*

Problem #5: Duna Numerals (20 marks)

Below are some numbers written in the Duna Language and their translations in the Hindu-Arabic numeral system, arranged in no particular order.

Malura yaba, Konane, Pou pou malura-da yaba, Malura-da rabagia,
Raga raga konane, Konane konane malura-da rabagia, Malura-da yaba,
Rebu rebu rabagia, Malura yaba-da pou, Yaba, Malura

Translations: 2, 10, 12, 16, 14, 24, 28, 36

Assignment 1
Find the correct correspondences between the Duna numbers and their translations.

Assignment 2
Write in Duna: 70, 112, 196

Duna belongs to the Trans-New Guinea group of languages and is spoken in Papua New Guinea by around 20000 speakers.

- *Sagar Sarda* and *Zarana Sarda*

Problem 4: Divehi Numerals (15 marks)

Given below are some traditional numbers in *Divehi* with their *Hindu-Arabic* numeral translations, not necessarily in order.

faheh, dholhas dhihaeh, fassihi dhihaeh, fanas, dholhas faheh, hatheh,
thin dholhas faheh, thineh, dholhas hatheh, fassihi, dholhas hathareh,
hatheh dholhas hathareh

7, 19, 48, 88, 17, 3, 24, 16, 22, 5, 34, 41

Assignment 1:

Match the numbers to their corresponding traditional *Divehi translations*.

Assignment 2:

Translate the following numbers to traditional *Divehi* numerals. If you are unable to translate a number based on the given data, write "*NP*" in your answer booklet, and explain your reasoning in your solution.

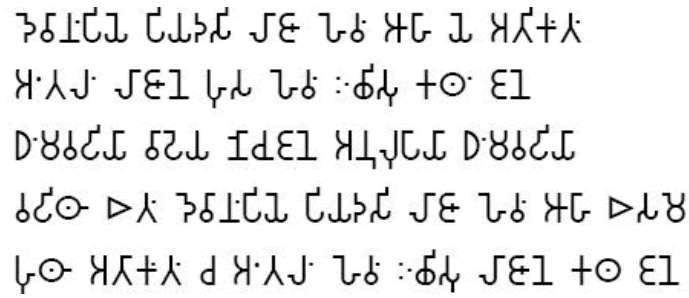
a. 15 b. 53 c. 67 d. 76

Divehi, also called *Maldivian*, is an *Indo-Aryan* language predominantly spoken by about 350,000 people in the Maldives, where it is the national language. *Divehi* is a descendant of *Maharashtri Prakrit* and is closely related to *Marathi*, *Konkani* and *Sinhalese*, but not mutually intelligible with them. The traditional *Divehi* number system was used until the middle of the 19th century, when the number system was restored to a modern decimal system.

- Sesh Sadasivam

Problem #6: Deciphering the Brahmi Script (20 marks)

An archeologist found a rock inscription from the 3rd century BC created by the Great *Mauryan* Emperor *Ashoka*. She examined the first five lines of this Ashokan edict, reproduced below:



She immediately figured out that it is in *Prakrit* – an ancient Indian language and is written using the *Brahmi* script. By comparing with other *Brahmi* inscriptions from the same time period, the archeologist could decipher the first line of this edict, which is:

devānaṁpiye piyadasi lājā hevaṁ āhā ye atikaṁtaṁ

The archeologist could not decipher the other lines completely, but she made a list of seven *Prakrit* words which she believed were present in the edict:

etaṁ, pari, nocujane, ca, esame, ichisu, dhaṁmavaḍhiyā

However, later it was found out that one of these seven words is actually not present in these five lines.

A line over a vowel indicates a long vowel. A dot under ***ṭ*** or ***ḍ*** indicates a retroflex sound (like in Hindi ***ṭamāṭar*** and ***ḍamru***), and is different from ***t*** and ***d*** without a dot, which indicates a dental sound (like in Hindi ***tum*** and ***desh***). The letter ***ṁ*** indicates nasal sounds.

Brahmi was an ancient Indian script. It most famously features in edicts of the Mauryan Emperor Ashoka. In modern times, it was deciphered by James Prinsep, a British archaeologist, in 1837.

Assignment 1:

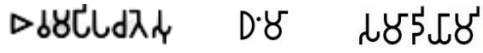
Find the word that is not present.

Assignment 2:

Match the remaining words from the list to the words in the *Brahmi* text.

Assignment 3:

Write down the following *Prakrit* words in English using the transcription scheme used here (see the gray box for details of transcription).



– Yash Sinha

Problem no 2. Japanese Days of the Week

The Japanese days of week are linked to the 7 luminaries/forces of nature. The week starts from Sunday and the 1st two days are linked with the Sun and The Moon, and the remaining 5 are linked to Fire, Water, Woods, Wealth, and Earth respectively. Based on the following clues, answer the question(s) that follow:

Nichibotsu – Sunset
Mokuhanga – Wood blockprint
Kaji – Fire event
Suimen – Water surface
Suion – Water temperature
Donabe – Earthen pot
Kin – Gold
Dochi – Land
Mokuyoubi – Thursday
Getsumen – Moon's surface

Find the Japanese words for the following days of the week:

Tuesday

Saturday

Monday

Friday

Sunday

Wednesday

Problem No. 1 (20 Points):

Basque

Consider the following Basque[1] words and their **unordered** English translations:

igogailu, artzain, lantegi, lantalde, bizitegi, taldekide, erizain, garbigailu, ikastalde, bizikide, garbitegi, ikaskide, lankide, eritegi, artalde

classmate, neighbour, sheep flock, crew, elevator, clinic, factory, nurse, home, shepherd, washing house, colleague, washing machine, team member, class (students)

Assignment 1. Pair up the Basque words with their correct English translations. Explain your solution.

Assignment 2. How is the word *artalde* different from the other Basque words given in the problem?

Assignment 3. Translate the word *sheepfold* into Basque, taking into account that it has the same feature as the word *artalde*.

-N. Zaika

[1] The Basque language is spoken by more than 660.000 people in the Basque Country (an autonomous community of Northern Spain). All existing hypotheses on the origin of Basque are controversial.

Problem No. 4 (15 Points):

Coptic

Consider corresponding words from two Coptic dialects — Sahidic and Bohairic (written out in Latin transcription) — and their English translations. The stressed vowels are marked in all polysyllabic words, excluding the last one:

	Sahidic	Bohairic	Translation
1	cop	čʰop	foot
2	bl̥lé	bellé	blind
3	pĺ̥ce	pʰélči	tatters
4	kúpr̥	kʰúper	henna
5	smótn̥	smóten	it is easy
6	petčí	petčʰí	receiving
7	čóftn̥	čʰóften	over the edge
8	pisabé	pisabé	that wise man
9	capíče	čapʰíči	capitha
10	čenepőr	čenepʰőr	roof
11	tetn̥ríme	tetenrími	you cry
12	čátfe	čʰátfi	reptile
13	čm̥péh	?	apple
14	anók	?	I
15	cépe	?	cloud
16	tápn̥	?	cummin
17	?	amre	baker

Assignment. Fill the gaps and mark the stress in the word *amre*. Explain your solution.

Note. *c* is pronounced similarly to English *k* as in *keep*, *č* is pronounced similarly to English *ch* as in *cheap*, the diacritic symbol ̥ under a consonant marks that this consonant is syllabic, the diacritic symbol ʰ after a consonant marks that it is pronounced in a specific (aspirated) way. *ô* and *ê* are specific vowels of the Coptic language. Capitha is a measure of volume (approximately 2,3 l).

[1] Coptic is a descendant of the Egyptian language. It is used in church services by Egyptian Christians.

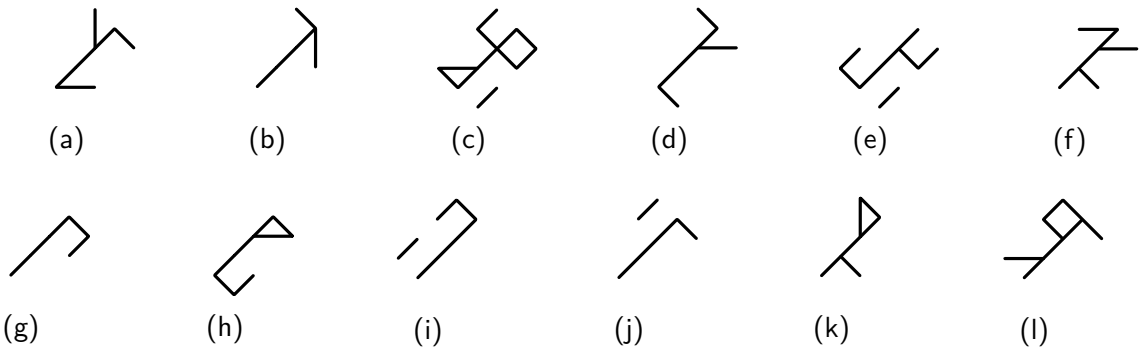
- S. Malyshev

Cistercian Numerals

A long time ago, hundreds of monks lived in exile, and needed to communicate with their friends on the outside. They had secret channels passing through trade routes, but there was always the risk of interception. To protect their information, they encoded their messages using a cipher.

You are a friend of the community, and new to the practice of trade, so you do not properly know the cipher. You are carrying a message that needs to go across a trade route. Unfortunately, you believe there will be guards at the next crossroads, and you don't want to be caught carrying secrets. Before crossing, you must decipher the message and memorize it in your head, destroying all evidence of such communication.

Luckily, you have mostly figured out the cipher, except for the numbers. Below are some numbers that appear in the message written using the ✠ Cistercian numerals ✠ notation[1]. (The numbers are written in no particular order.)



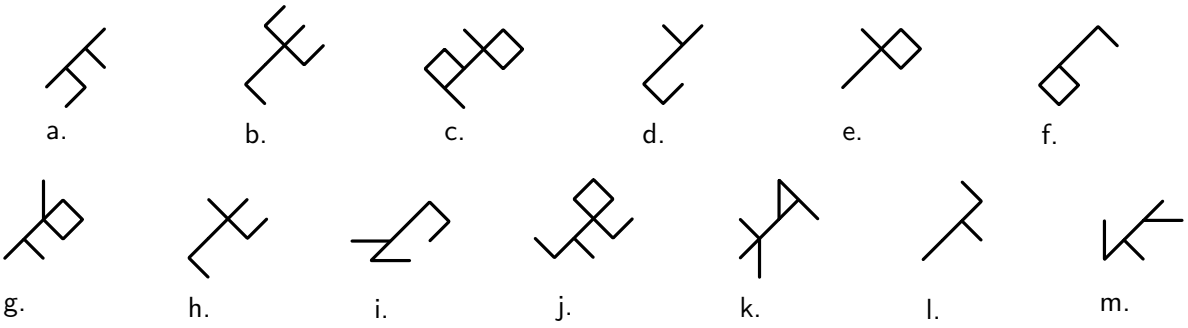
You are confident that their counterparts are the following numbers (written in ascending order):

7, 13, 61, 114, 234, 250, 341, 705, 4091, 5689, 6070, 7608

Assignment 1: Your task is to match the ✠ Cistercian numerals ✠ with their counterparts.

[1]The medieval Cistercian numerals, or "ciphers" in nineteenth-century parlance, were developed by the Cistercian monastic order in the early thirteenth century at about the time that Arabic numerals were introduced to northwestern Europe.

Assignment 2: Now back from your journey, you must relay a message from the outside to the community in exile. How would you write down today's date (249/12/1298) on the message in the ✠ Cistercian numerals ✠ notation? (There is a typo in the date, but encode it as-is, regardless). Pick three numbers, one each for a date, a month, and a year, from the following collection and indicate which is which (e.g., date: x., month: y., year: z.).



– Aalok Sathe

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.

2. Circles and Dots (20 marks)

Problem designed by Anindya Sharma and Monojit Choudhury

Inspired by Sherlock Holmes in "The Adventure of the Dancing Men", Golu and his best friend Hammie decided to invent a code language for writing letters to each other. How cool, they thought, it would be to have a language that only they understood! However, they soon realized that drawing the 26 dancing men figures, a different one for each letter of the English alphabet, was not their cup of tea. Golu then had this brilliant idea that they could instead use fewer and simpler symbols: a dot (•), a circle (○) and a circle with a dot (⊙), and each English letter could be defined by a unique sequence of these symbols. He came up with the following scheme for encoding the first 7 letters of the alphabet:

Scheme 1 – A: •○○⊙, B: •○⊙, C: ••○, D: ○○⊙, E: ○•, F: ○○, G: ⊙

In this scheme, they would encode the word CAB as: ••○•○○⊙•○⊙

Hammie's elder brother Shan, who was studying computer science, happened to pass by. He saw Golu's code and immediately pointed out that this coding *scheme* won't work. They would be in trouble, though he left it for Golu and Hammie to figure out why it won't work and what scheme would work. Not sure of the solution, Golu and Hammie thought of designing 4 more coding schemes:

Scheme 2 – A: ○⊙, B: ⊙•, C: ⊙⊙, D: ○○○, E: ⊙•⊙, F: ○○⊙•, G: ○○⊙

Scheme 3 – A: •, B: ○, C: ⊙, D: ••, E: •○, F: •⊙, G: ○•

Scheme 4 – A: ○•, B: ○○, C: ⊙⊙, D: ••, E: ⊙○, F: ⊙•, G: ○⊙

Scheme 5 – A: ⊙•○, B: ⊙•⊙, C: ⊙○⊙, D: ⊙•○⊙, E: ⊙••, F: ⊙○○, G: ○○

They went to Shan, who now said that schemes 2 and 4 were fine; but schemes 3 and 5 have the same problem as scheme 1, and should not be used. Golu looked at the 5 schemes carefully and exclaimed: "Eureka! I know what the problem is with schemes 1, 3 and 5. I feel so stupid to have come up with such schemes!"

Shan was not sure though if Golu really understood the problem. He showed them 3 more schemes and asked them to figure out which of these would work.

Scheme 6 – A: ••, B: •○, C: •○⊙, D: ○•, E: ○○, F: ○⊙, G: ••○

Scheme 7 – A: ⊙⊙, B: ⊙○, C: ⊙•, D: ○•, E: ○○, F: ○⊙, G •

Scheme 8 – A: •○○, B: •○⊙⊙○, C: ○⊙⊙○⊙, D: ○⊙⊙•, E: ○•••••, F: ○⊙○•, G: •○•○

Assignment 1: What do you think is the problem with the schemes 1, 3 and 5, that schemes 2 and 4 do not have?

Assignment 2 [Answer Booklet]: Among the schemes 6, 7 and 8, which one(s) has/have the same problem as 1, 3 and 5?

Golu had indeed got his answers right. But now that Hammie and he had to extend their scheme for the remaining 19 letters, they needed a systematic process for designing coding schemes which would avoid the problem found in schemes 1, 3 and 5. Shan gave them a hint. He said "of the first 5 schemes, scheme 4 is special because none of the codes is a prefix of the code for another letter." Such codes are called *prefix codes*. Scheme 3, for example, is not a prefix code because the code for A is a prefix of the code for D and E; similarly the code for B is a prefix of the code G.

Assignment 3 [Answer Booklet]**:** Is it true that any prefix code will avoid the problem that schemes 1, 3 and 5 have? Explain why or why not.

Assignment 4 [Answer Booklet]: Is it true that for a code to be useful, it must be a prefix code? Explain why or why not.

Notes: This problem is based on *Coding theory* which is the study of the properties of coding schemes and their fitness for a specific application. Codes are used for data compression, cryptography, error-correction, and networking. Codes are studied by various scientific disciplines—such as information theory, electrical engineering, mathematics, linguistics, and computer science. In 1948, scientist Claude Shannon published an article - "A Mathematical Theory of Communication" which laid the foundations of coding theory. Marcel J. E. Golay of *Golay code* fame and Richard Hamming who lent his name to *Hamming codes, Hamming windows, Hamming numbers,* and *Hamming distance* were some of the pioneers of this field.

(A) PT PRP WDPREVTAG VF it

(B) PP **(11)** **(12)** DT since

(C) PP PT WDPREVTAG VF WDNEXTTAG PP

(D) NNS VF **(13)** NNP

(E) VF NN WDPREVTAG **(14)**

(F) NNS **(15)** WDNEXTTAG PT

Assignment 1: (12 marks)

Fill in the omitted entries numbered (1)–(15). It may be helpful to know that rule (C) affects only one word and rule (E) does not affect sentence (c).

Assignment 2: (3 marks)

What will Brill's Tagger output for this sentence in both steps?

"He books it for another show."

– Monojit Choudhury

Problem 2: Chickasaw Stress (17 marks)

Here are some words in Chickasaw and their meanings. The words are in a phonetic notation where the exact sounds are not important, but:

- o, a, i are vowels, and : after a vowel indicates that the vowel is long
- ʃ, ɬ, ʔ, and the other letters are consonants
- ˈ at the beginning of a syllable indicates primary stress (i.e. emphasis) on that syllable, and ˌ indicates secondary stress

Word	Meaning	Word	Meaning
tʃoˈkaːˌno	fly	ˌmaʃˈkoːˌkiʔ	(name of a tribe)
ˌlokˈtʃok	mud	ˌokˌfokˈkol	(type of snail)
ˈaːˌtʃomˌpaʔ	local store	taˈlaːˌnomˌpaʔ	telephone
iˌbiɬˈkan	snot, mucus	ˈnaːɬtoˌkaʔ	policeman
ˌʃanˈtiʔ	rat	aˈboːkoˌʃiʔ	river
ˈsaːɬkoˌna	earthworm	noˌtakˈfa	jaw
ˌtʃonˈkaʃ	heart	tʃiˌkaʃˈʃaʔ	Chickasaw
faˈlaːt	crow	ˌokˈtʃaːˌlinˌtʃiʔ	saviour

Assignment 1: (7 marks)

If you are given a new Chickasaw word, how would you identify the syllables? Then, what steps would you take to determine which syllables get primary stress and which ones get secondary stress?

Assignment 2: (10 marks)

Here are some more words in Chickasaw. Mark the stress with ˈ and ˌ at the beginning of the syllables that carry primary and secondary stress, respectively.

	Word	**Meaning**
(a)	taʔossa:pontaʔ	finance company
(b)	ʃimmano:liʔ	(name of a tribe)
(c)	kanannak	(type of lizard)
(d)	intikba:t	sibling
(e)	okta:k	prairie

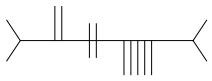
– Saujas Vaduguru

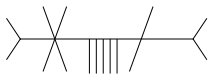
Problem 3: Ire-lined (11 marks)

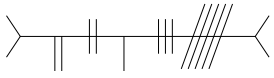
Ogham is an Early Medieval writing system used for the Ancient (Early) Irish language and later the Old Irish language. There are roughly four hundred surviving inscriptions using Ogham (the so-called "orthodox inscriptions") on stone monuments throughout Ireland and western Britain.

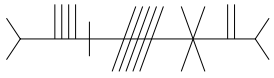
Assignment 1: (4 marks)

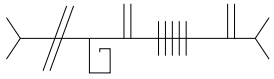
Match these Ogham (O) transcriptions to their Ancient Irish (AI) counterparts and their English (E) meanings. (Note: There is one fewer O option than there are AI and E options. The extra AI and E match up to each other, and to one O option from Assignment 2.)

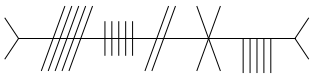
A. 

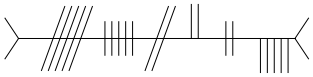
B. 

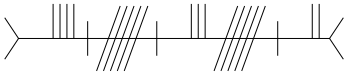
C. 

D. 

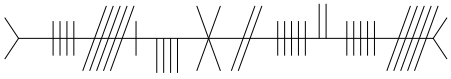
E. 

F. 

G. 

H. 

I. 

J. 

Problem #3: Cherokee Numerals (15 marks)

Below are some numbers written in the *Cherokee* language arranged in no particular order, and their corresponding translations in the Hindu-Arabic numeral system, arranged in ascending order.

sudalisgo hisgi, sowo, hisgadu, hisgi, galiquogi, galiquasgohi, talisgohi, talidu, nvgisgo sawo, tsoigo tsoi, galiquadu, hisgisgohi, galiquasgo sudali, daladu.

Translations: 1, 5, 7, 12, 15, 16, 17, 20, 33, 41, 50, 65, 70, 76

Assignment 1
Find the correct correspondences between the *Cherokee* numbers and their translations.

Assignment 2
Write in *Cherokee*: 36, 55, 6, 60, 21

Assignment 3
Write in numbers: **talisgo sudali, sudalisgo sudali, hisgisgo nvgis**

Cherokee is the Native American Indian *Iroquoian* language spoken by the *Cherokee* people. There are approximate 10 to 20 thousand Cherokee speakers in the world today, most of whom live in the Oklahoma and the North Carolina states of the USA.
v is a nasal vowel.

- Prachatos Mitra

2. Chamorro Months (20 marks)

Problem designed by Monojit Choudhury

After finishing her undergraduate degree in mathematics, Ms. Vibhashini wanted to explore the world. She took up a job of teaching English in a primary school in the island of Guam located in the Pacific Ocean. One day, in the class, she asked her students to write down the names of the months in English. However, she was surprised to see that 10 year old Tasi, who is a native of the island and speaks Chamorro at home, came up with a list of 23 months:

Abrit	*Hulio*	*Maimu'*	*Oktubri*	*Umagahaf*
Agosto	*Huño*	*Makmamao*	*Semu*	*Umatalaf*
Disembre	*Ineru*	*Måtsu*	*Sumongsong*	*Umayanggan*
Fananaf	*Lumamlam*	*Måyu*	*Tenhos*	
Fangualu	*Lumuhu*	*Nubembre*	*Tumaiguini*	

Vibhashini asked Tasi what were these. None of them were the names of the months she was expecting. Tasi told her that his grandmother tells him that there are 13 months and he learnt the names of those months. But then his parents use some other month names, some of which he remembers as well. Tasi was confused as to which month names his teacher wanted. Therefore, he wrote all the 13 names that his grandmother taught him, as well as 10 more month names that he could remember that his parents use. But he'd mixed all of them up.

Vibhashini looked at the names again, and smiled at Tasi. She knew what he was talking about and which months he missed out. She helped Tasi remember those missing names, even though she did not know Chamorro. Can you separate out the 13 month names used by Tasi's grandmother from the 10 that his parents use?

Assignment 1 [Answer-Booklet]: Identify the 10 months that Tasi has heard from his parents and write their English equivalents?

Assignment 2 [Answer-Booklet]: What are the two missing month names in Chamorro? What are their English equivalents? [No explanations needed for parts 1 and 2]

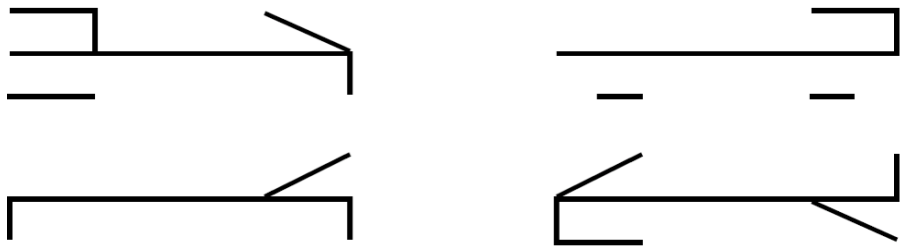
Assignment 3: Why do you think Tasi's grandmother and his parents use a different set of names for the months?

Note: Chamorro is an Austronesian language spoken by about 47,000 people who are the indigenous people of Guam and the Northern Mariana Islands, US territories. u' and $å$ are vowels, $ñ$ is a consonant.

Problem No. 2 (20 Points):

Numbers

Consider the following four-digit numbers written out in the code proposed by a Dutch scholar Jan van Bronckhorst (1494–1570) in 1539:

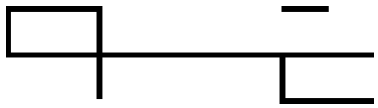


Consider the same numbers **in another order** written out in

figures: 1368, 1410, 4173, 5750

Assignment 1. Match the corresponding numbers. Explain your

solution. Assignment 2. Write down the following number in figures:



Assignment 3. Write down the following number using the code of Jan van Bronckhorst:

7942

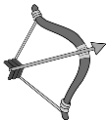
-S. Burlak

Problem #3: Bow or Onion? (15 marks)

Words often have multiple meanings. For example, the English word ***crane*** can either mean 'a bird' or `a machine used for lifting heavy objects`. Similarly, the Russian word ***лук*** **(**pronounced as **luk)** has two meanings: ***onion*** and ***bow*** (a weapon). In the first paragraph below the word **luk** means ***onion*** whereas in the second paragraph it means ***bow***.



shalot i **luk** (***onion***) ochen tesno svjazannye ovoschi. oba imejut lukovichnye rastenija. v vkusa, shalot bolee ostryj i, sledovatelno, ne edjat v kolichestvakh.



bak gorazdo bolee prodvinutyj, oruzie, chem **luk** (***bow***). vmesto togo, chtoby streljali iz lukov ona ispolzuetsja, chtoby streljat snarjadami. eti snarjady ispolzujutsja v sovremennoj vojne, chtoby ubit vragov.

For each of the following paragraphs containing the word **luk**, identify whether it means ***onion*** or ***bow***? Explain how you arrived at your conclusions. For ease of reading, the Cyrillic letters used for writing Russian have been mapped to similar sounding Latin letters.

1. pistolet javljaetsja gorazdo bolee sovremennyj oruzie, chem luk. vmesto togo, chtoby strely iz lukov on ispolzuetsja dlja metanija puljami. on ispolzuetsja v sovremennoj okhoty ubit zivotnykh.

2. chesnok i luk tesno svjazannye ovoschi . oni oba imejut lukovichnye rastenie. v vkusa, chesnok bolee ostryj i, sledovatelno, ne edjat v bolshikh kolichestvakh.

3. okhotnikov ispolzovat luk iz dereva dlja okhoty na razlichnykh zivotnykh. oni chasto imel navyki v strelbe iz luka i praktika ego s samogo rannego vozrasta. oni ispolzujut luk, metanija strely i ubit nichego ne podozrevajuschikh zertv.

4. luk luchshe kultiviruetsja v plodorodnykh pochvakh, kotorye khorosho drenirovannye. v otlichie ot mnogikh drugikh ovoschej, luk trebujut vysokogo urovnja pitatelnykh veschestv v pochve. rastenie trebuet 11-12 chasov solnechnogo sveta.

5. luk soderzit paru izognutykh uprugikh konechnostej iz dereva. obe konechnosti soedineny stroki. okhotniki specialistam v strelbe iz luka tjanet stroku nazad, chtoby metanija.

6. luk ispolzuetsja kak ovosch i javljaetsja naibolee shiroko kultiviruemykh i edjat vidov. eto naibolee chasto raz v dva goda ili mnogoletnee rastenie. ego listja golubovato - zelenogo cveta.

7. luk i chesnok obychno ispolzujutsja v kachestve prigotovlenija pischi ingredienta dlja povyshenija vkus indijskikh karri. on ispolzuetsja v kachestve zagustitelja dlja karri. on takze edjat kak salat vmeste s drugimi ovoschami.

8. luki i strely byli shiroko ispolzuetsja v primitivnoj vojny. v sovremennoj vojne, oni byli zameneny orudij i raket. khotja strelba iz luka bolshe ne trebuetsja v sovremennoj vojne ne do sikh por praktika kak vid sporta.

9. luk i strely predstavljaet soboj sistemu snarjad oruzie. strelba iz luka eto iskusstvo, praktika ili iskusstvo ispolzovanii luka dlja metanija strel. eto ochen primitivnoj sistema ispolzuetsja okhotnikov i peschernogo cheloveka.

10. luk marinovannyj v uksuse edjat v kachestve zakuski. ego listja inogda ispolzuet dlja podgotovki indijskikh karri. luk takze mozno upotrebljat v pischu syrye otlichie chesnok, kotoryj takze javljaetsja ovosch prinadlezaschikh k odnomu vidu.

Choosing the correct meaning of a word in a given context (sentence or paragraph) is an important task which must be solved for developing language processing and understanding software such as automatic translation systems. This task is known as *Word sense disambiguation*. The cues that a computer uses to solve this task is probably similar to what you will be using to solve this problem!

- *Mitesh Khapra*

5. Aryabhat's Enigma (20 marks)

Problem designed by Apoorva Bhagwat and Angikar Ghoshal

Lenny was studying number theory from an old Sanskrit tome, when he came across the following curious shloka (left, transcribed using Latin script from the original Devanagari script):

tukā nātha kabīra ete ādikavayah |	*Tuka*, *Nath* and *Kabir* were the original poets.
nāmā gadimā kintu dwijakavīdvayah ||	*Nama* and *Gadima* were both born twice.
gula ravi thākūra gāliba trimūrtī |	*Gulzar*, *Ravi Thakur* and *Galib* were a trio.
bahiṇā ekamevā caturastrā kavayitrī ||	*Bahina* was the sole versatile poet.

Lenny soon realized that the shloka was referring to several Indian poets: Tukaram, Eknath, Kabir, Namdeo, G. D. Madgulkar (*gadimā*), Gulzar, Ravindranath Thakur, Mirza Galib and Bahinabai Chaudhari. Being a good student of Sanskrit, Lenny promptly translated the shloka, and came up with this slightly nonsensical translation (shown in the right above).

Realizing that there was something fishy about this shloka, Lenny took it to Prof Arya Bhat, and asked him what the shloka meant. "Ah, a classic Aryabhata numeration shloka", said the professor, pulling out a piece of paper from his drawer. "Each poet represents an integer with a different number of distinct prime factors", he said.

Poet	**Integer**	**Number of distinct prime factors**	**Poet**	**Integer**	**Number of distinct prime factors**
tukā	160001	1	*gula*	30050	3
nātha	37	1	*ravi*	6040	3
kabīra	2341	1	*thākūra*	10052	3
nāmā	45	2	*gāliba*	5026	3
gadimā	1828	2	*bahiṇā*	10038	4

"The numbers are represented by the words according to a scheme called the *Aryabhata numeration*. I do not exactly remember how the scheme works, but I'm confident the numbers on this paper are correct. Perhaps you can try to reverse-engineer the scheme!" While you do not need to know the Devanagari script to decipher the scheme (the Latin transcriptions have all the information you need), the organization of the letters in the Devanagari alphabet might be useful:

Vowels: *a ā i ī u ū ṛ ṝ ḷ ḹ e ai o au*

Consonants: *k kh g gh ṅ c ch j jh ñ ṭ ṭh ḍ ḍh ṇ t th d dh n p ph b bh m y r l v ś ṣ s h*

Assignment 1 [Answer-Booklet]: Write down the numbers corresponding to the following words:

a) *jāveda*, b) *kāli*, c) *tulasī*, d) *dāsa*, e) *sāhira*

Assignment 2 [Answer-Booklet]: Write down words representing the following numbers (if there are multiple valid representations, any single one is acceptable)

f) 230019, g) 7070, h) 21000017

Assignment 3: Explain the Aryabhata numeration scheme.

Problem 5: Alien Protein Codes (13 marks)

Dr. Muzabique figured out how to manipulate six alien proteins to do three functions: construct, cut, and pack. The alien genetic code used to manipulate the proteins was found to have 6 "alphabet letters": `A`, `T`, `G`, `C`, `D`, `N`. Only he knew the top-secret algorithm, and two weeks ago, he was mysteriously found dead in his apartment.

The top-secret project is now in a fund crunch but the government decides to give it one more chance if the algorithm can be decoded. They send you the only page that was recovered from Dr. Muzabique's notebook, with one set of input instructions and output genetic codes, and ask you to decode the algorithm:

"Today was a great breakthrough! I found that large and small proteins inherently behave in different ways. The way they operate under different cellular functions..."

Input: `Celebi, construct Articuno`
Output: `CNDATACACCNDTGTATTCGGGATGCNDTGCCCACCCCNNNCNDCTTTTCCACTNANCND`

Input: `Articuno, construct and cut Terrakion`
Output: `CNDTGTATTCGGGATGCNDACTGCAGGGACTCCNDGCATGACATCNNNNNNCNDTGCCCACCC CNNNCND`

Input: `Terrakion, construct and pack Shaymin`
Output: `CNDACTGCAGGGACTCCNDCAGGCTCNDGCATGACATCNNNNANCNDCTTTTCCACTNNNCND`

Input: `Cressela, construct and cut and pack Azelf`
Output: `CNDCCATATGATGCGACNDGTTGATCNDGCATGACATCNNNNANCNDTGCCCACCCCNANCND CTTTTCCACTNNNCND`

Input: `Celebi, cut Cressela`
Output: `CNDATACACCNDCCATATGATGCGACNDGCATGACATCNANNNNCNDCTTTTCCACTNANCND`

Input: `Shaymin, pack and construct Azelf`
Output: `CNDCAGGCTCNDGTTGATCNDTGCCCACCCCNANCND`

Input: `Celebi, construct and cut and pack Cressela`
Output: `CNDATACACCNDCCATATGATGCGACND`

Input: `Terrakion, pack Articuno`
Output: `CNDACTGCAGGGACTCCNDTGTATTCGGGATGCNDCTTTTCCACTNNNCND`

Assignment 1: (13 marks)

Preserve the legacy of Dr. Muzabique! Explain how the algorithm converts input commands into output genetic codes.

– Anshuman Swain

1. Akamu's Table (20 marks)

Adapted from "Compilation of Russian Olympiad problems between 1965 and 1975" by Vladimir Belikov, Elena Muravenko, Alekseev Mikhail Egorovic

"Did you know that the Hawaiian language and the Maori language spoken by the Maori people of your country are related?" said Akamu, the tourist guide, to Tia, the tourist from New Zealand, who was visiting Hawaii for the first time. "That's impossible!" exclaimed Tia. "The two archipelagos are separated by 7500 km of water. How could two languages spoken in islands so far apart could be similar?" Akamu smiled. "Well, it is not only these two languages. There are at least 30 other languages spread across the islands of the Pacific Ocean that evolved from the same common ancestor. Linguists call this the *Polynesian family of languages*". He took out a piece of paper and wrote down the words for the cardinal numbers (first, second, third etc.) in 5 Polynesian languages that he could recall. He handed it over to Tia and said "You don't have to believe me; see it for yourself. How would a set of unrelated languages have so similar words for the cardinal numbers?"

Tia looked at Akamu's table with utmost surprise (' and wh are specific consonants):

Language	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
Hawaiian	*kahi*	*lua*	***(a)***	*ha*	*lima*	*ono*	*hiku*	*walu*	***(b)***
Maori	*tahi*	*rua*	*toru*	*wha*	***(c)***	*ono*	*whitu*	*waru*	*iwa*
Nuku Hiva	*tahi*	***(d)***	*to'u*	*ha*	***(e)***	*ono*	***(f)***	*va'u*	***(g)***
Rarotonga	*ta'i*	***(h)***	***(i)***	*'a*	*rima*	*ono*	*'itu*	*varu*	*iva*
Samoa	*tasi*	*lua*	***(j)***	***(k)***	*lima*	*ono*	*fitu*	***(l)***	*iva*

It was not hard to see the pattern. Akamu had forgotten some of the words – the gray cells in the table. Tia could fill in the gaps simply by studying the pattern. She showed Akamu her guesses, and Akamu exclaimed "Woah! You got everything right. When did you learn all these languages?" "It's actually simple", said Tia.

Assignment 1 [Answer Booklet]: Guess the 12 missing words.

Assignment 2: Explain the rules that you used to convert words from one language to another. Here is one example rule (as well as a hint) for you: 'the consonant *l* in Hawaiian is transformed to *r* in Maori'.

Problem no 4. Tangkhul Kinship Terms

Here is a partial tree diagram representing the family of Achui, a Tangkhul-Naga speaker female. The tree shows Achui's generation and her parents. Achui is explaining the family tree to Tsimik, her son. Accordingly, Tsimik has to complete the tree by adding his generation in the tree. There are five people to be added: Tsimik, G, H, I and J. Can you help him?

(Note: In Achui's family, Achui, X, E, C, G, H, I are females, while Tsimik, Y, B, D, F and J are males).



Achui =D B

=E C =F [?] [?] [?] [?] [?]

1. X iwi aavu.
2. i Cwi aachon.
3. C iwi atsarvu.
4. C Ywi naaongalaavu.
5. B Xwi naaomayarri.
6. Y iwi aavaa.
7. X Dwi aanyi.
8. Y Fwi aawo.
9. i nawi aavu.
10. i Hwi aavu.
11. H Dwi irri.
12. G nawi atsarvu.
13. H Iwi aachon.
14. I Cwi naaongalaavu.
15. na Bwi aarri.
16. J Cwi irri.
17. D Iwi aavaa.
18. B nawi aawo.
19. E Hwi aanyi.
20. J iwi irri.

B] Fill in the blanks.

1. G iwi_____.
2. D Hwi_____.
3. Y Ewi_____.
4. C Tsimikwi_____.
5. Tsimik Fwi_____.

Problem #2: Abracadabra (15 marks)

Prof Lewos Caril, a computational linguist, has designed an artificial language *Abracadabra* that uses only five letters **A, B, C, D** and **R.** A peculiar characteristic of this language is that words are written one after another without any space or punctuation marks in between. In fact, the name of the language is actually ***Abra cad abra*** meaning "*word after word*", but it is written as ***Abracadabra***. It could also be decomposed into another sequence of valid *Abracadabra* words: ***Ab rac adabra*** which translates to "*the of night*" and therefore, is not really a meaningful decomposition.

Consider the following Abracadabra sentence:

abbaradracradababadarba

Here is the dictionary of all the *Abracadabra* words along with their meanings and parts-of-speech in parenthesis (n= noun, v = verb, a = adjective or adverb, p = preposition):

ab	the	***car***	before (p)
abra	word (n)	***cra***	sun (n)
adabra	night (n)	***crada***	sky (n)
adar	teacher (n)	***dabab***	enchant (v)
adrac	student (n)	***draba***	dance (v)
arba	red (a)	***dara***	never (a)
ba	always (a)	***rac***	of (p)
babad	glow (v)	***racrad***	grow (v)
bar	good (a)	***rad***	blue (a)
baradra	morning (n)	***radab***	respect (v)
cad	after (p)		

Assignment 1:
Write down all possible ways in which this sentence can be decomposed into sequences of valid *Abracadabra* words.

Assignment 2:

Not all of these decompositions are grammatically correct or meaningful sentences. Which of these decompositions do you think are correct and meaningful? Give their possible English translations. Beware that Abracadabra grammar can be different from English grammar.

Prof Lewos Caril also designed a computer program to automatically segment an Abracadabra sentence, which works as follows. It scans the sentence from left to right and identifies the longest valid Abracadabra word that matches the sentence from the left. This word is segmented from the sentence, and the remaining part of the sentence is then decomposed likewise by iteratively removing the longest matching valid word from the left. At any point of time, if no matching valid word can be found, then the program stops and declares that the sentence *cannot be decomposed*. Otherwise, the program gives as output, one of the possible decompositions of the sentence.

Assignment 3:

What will be the output of Prof Caril's program for the above sentence?

Assignment 4:

Is it possible that a sentence has a valid decomposition but Prof Caril's program declares that it cannot be decomposed? If *yes*, construct such an example sentence and clearly show the valid decomposition as well as the invalid partial decomposition that Prof Caril's program produces. If your answer is *no*, then explain why not?

In many languages, such as Thai and Chinese, words are not separated by spaces or punctuation marks. Word segmentation, the process of breaking a sentence into words, is an important challenge that a computer needs to solve before processing these types of languages.

– Monojit Choudhury

5. A Rose by another Name (20 marks)

Problem designed by Abhishek Dedhe

You must have heard about chemical formulas, but did you know that there are also *floral formulas* for representing the structure of flowers? For example, the floral formula of *China Rose* is:

$$\text{Br. Brl. } \Phi \text{ K}(5) \text{ C}_5 \text{ A}(\infty) \text{ G}(5)$$

Don't worry if that looks bizarre. Not all flowers have such complicated formulas. In fact, there is a systematic way in which the structure of a flower (for example, whether it is symmetrical, how many petals does it have, how are the petals connected to each other, and so on) can be represented in terms of a *floral diagram*. Since drawing a detailed diagram can take time, botanists have invented a clever way of representing the diagrams in terms of floral formulas.

Given below are some floral diagrams and their corresponding floral formulas.



$K_4 C_4 A_4 G_1$



$K_0 C_5 A_{10} G_1$



$K_{(4)+1} C_5 A_{4+1} G_5$



$K_4 C_{(2+2)} A_{4+2+2} G_4$

Assignment 1: Explain the rules of obtaining the *floral formula* from a *floral diagram*.

Assignment 2 [Answer-booklet]: Given below are three floral diagrams. Write their corresponding floral formulae.



(a)



(b)



(c)

Assignment 3: Explain the problem(s) which would arise when trying to draw the floral diagram of this floral formula: $K_{(6)}C_{(2+2)}A_4G_2$ How might you resolve this issue?

Note: The data presented in this problem has been slightly simplified.

4. A Dozen Miskito Birthdays (20 marks)

Problem designed by Shinjini Ghosh

Sumu has 12 friends; let's denote them by the letters from A to L. One day Sumu's friends decided to play a prank on her. They swapped their names, photographs and all other information on their Facebook profiles randomly among themselves. Sumu had no way to recognize who was who. However, her friends did not change one information – their birthdays. It was a clue that Sumu could use to identify her friends, if she remembered all of their birthdays. Fortunately, Sumu did remember her friends' birthdays. But to her surprise, her friends had listed their birthdays not in English, but in an unknown language called *Miskito*. Sumu listed down the birthdays of her 12 friends, A to L, and the Miskito dates mentioned in her friends profiles (in random order):

Dates in English	**Miskito Dates**
A. 10th December	1. Matawalsip pura kumi, Kuswa Kati
B. 6th March	2. Yumhpa, Siakwa Kati
C. 2nd October	3. Matawalsip pura wal, Wis Kati
D. 11th February	4. Matlal kahbi pura wal, Yahbra Kati
E. 7th May	5. Matlal kahbi, Kakamuk Kati
F. 12th September	6. Walwal, Lih wainhka Kati
G. 3rd January	7. Wal, Waupasa Kati
H. 9th August	8. Matawalsip, Trisu Kati
I. 8th November	9. Matlal kahbi pura yumhpa, Sikla Kati
J. 5th July	10. Kumi, Li Kati
K. 4th April	11. Matlal kahbi pura kumi, Lih Mairin Kati
L. 1st June	12. Matsip, Pastara Kati

Can you help Sumu to find out who is who by matching the Miskito dates to the English ones?

Assignment 1 [Answer-Booklet]**:** Match the Miskito dates with their English translations.

Assignment 2 [Answer-Booklet]: What do you think 'Kati' means literally?

Assignment 3 [Answer Booklet]**:** Sumu then decides to write her teachers' birthdays in Miskito. Help her to write the following dates in Miskito:

(a) 6th July, (b) 9th April, (c) 10th January, (d) 5th May, (e) 7th September

Note: Miskito is spoken by the Miskito people who live in Northeastern Nicaragua. There are approximately 150000 Miskito speakers.

Problem 3: Luganda (15 marks)

Given below are some words in *Luganda*, along with their *English* translations in random order:

kantu, muti, kagga, nbogo, bulawuzi, kawuzi, nbwa,
bulangiti, mugga, bayibuli, kabwa, muti

blanket, tree, dwarf, dog, stream, the Bible,
thread, buffalo, river, blouse, puppy, pole

Assignment 1:
Determine the correct correspondences.
(**Hint:** *muwuzi* → *rope*)

Assignment 2:
Given below are some more Luganda words along with their English translations in random order. Using the information that you got from Assignment 1, determine the correct correspondences.
(**Hint:** *mata* → milk)

mazzi, nkoko, npisi, mazzi, kakoko, kalifoomu

water, hen, chloroform, hyena, urine, chick

Luganda is the most widely spoken language in Uganda. It is spoken by about 5 million people. It belongs to the Bantu branch of the *Niger-Congo* language family.

- Abhishek Dedhe